Smart Traffic and Surveillance Management System

F26-095-EquiFlow

Project Team

Abdul Hadi Ayaz Abbasi	23I-0707
Muhammad Adeel	23I-0739
Marvan Shahid	23I-0800

Session 2023–2027

Supervised by

Dr. Suleman Khan

Co-Supervised by

Dr. Rizwan



FAST School of Computing

National University of Computer and Emerging Sciences
Islamabad, Pakistan

October, 2026

Contents

Students' Submission i

Anti-Plagiarism Statement iii

Authors' Declaration v

Abstract vii

Acknowledgments ix

Executive Summary xi

List of Abbreviations xiii

1 Introduction and Project Vision 1

1.1 Project Overview . . . 1
1.2 Problem Statement . . . 1
1.3 Problem Context and Motivation . . . 1
1.4 Project Vision . . . 2
1.5 Project Objectives . . . 2
1.6 Project Scope . . . 2
1.7 Project Modules . . . 3
1.8 Stakeholders and User Classes . . . 4
1.9 Success Criteria . . . 4
1.10 Constraints and Assumptions . . . 4
1.11 Ethical, Legal and DEI Considerations . . . 5

2 Research and Existing Solutions 7

2.1 Research Method / Approach . . . 7
2.2 Existing Solutions / Products . . . 7
2.3 Comparison of Existing Solutions . . . 8
2.4 Limitations and Gaps . . . 8
2.5 Proposed Solution . . . 9
2.6 Complex Computing Problem . . . 9
2.7 Technical Challenges . . . 9

2.8 Research Findings and Design Decisions 10

3 Proposal Feedback and Iteration Plan **11**
3.1 Proposal Feedback Summary 11
3.2 Proposal Action Register 11
3.3 Iteration Strategy 12
3.4 Iteration 1 Plan 12
3.5 Iteration 2 Plan 12
3.6 Work Breakdown and Responsibilities 13
3.7 Progress Against Plan 13
3.8 Delays, Deviations and Recovery Plan 13

4 Software Requirements **15**
4.1 User Interactions / Use Cases 15
4.2 Use Case Diagram 15
4.3 Detailed Use Cases 16
4.4 Functional Requirements 16
4.4.1 Perception and traffic-state module 16
4.4.2 Dynamic control and recovery module 17
4.5 Non-Functional Requirements 17
4.5.1 Reliability 18
4.5.2 Usability 18
4.5.3 Performance 18
4.5.4 Security 18
4.6 Acceptance Criteria 19
4.7 Requirements Traceability 19

5 System Design **21**
5.1 System Architecture 21
5.2 Architectural Components 21
5.3 Interfaces and Dependencies 22
5.4 Architecture Decisions and Rationale 23
5.5 Design Evolution 23
5.6 Domain Model 23
5.7 Data Design 25
5.8 Process / Behavior Design 25

6 Iteration 1: Design and Implementation **27**
6.1 Iteration Objectives 27
6.2 Selected Use Cases / Features 27
6.3 Design 27
6.4 Implementation 27
6.4.1 Implementation #1 – SUMO simulation toolkit 28

6.4.2 Implementation #2 – YOLO26 ambulance detection 29
6.4.3 Implementation #3 – Live ambulance detection dashboard 29
6.4.4 Implementation #4 – Repository organization 30
6.5 Completed Features / Module 30
6.6 Technical Implementation Details 31
6.7 Algorithms and Technical Decisions 31
6.8 Tools, Frameworks and Technologies 32
6.9 Demonstration Evidence 32
6.10 Iteration Outcome 32

7 Testing and Early Validation **33**
7.1 Testing Strategy 33
7.2 Test Levels and Techniques 33
7.3 Test Cases 34
7.4 Test Results 34
7.5 Acceptance Testing / Validation 37
7.6 Edge and Failure Cases 37
7.7 Defects and Corrective Actions 37
7.8 Current Validation Status 38

8 Project Management and Development Evidence **39**
8.1 Work Completed 39
8.2 Work Remaining 39
8.3 Individual Responsibilities 40
8.4 Contribution of Each Team Member 40
8.5 Repository and Version Control Evidence 40
8.6 Milestones and Commit Evidence 40
8.7 Tools and Tool-Assisted Development 41
8.8 External Components and Dependencies 41
8.9 Risks and Mitigation 41
8.10 Security and Privacy Considerations 41
8.11 Current Limitations 42

9 Remaining FYP Plan **43**
9.1 Remaining Scope 43
9.2 Remaining Modules and Features 43
9.3 Remaining Iterations 44
9.4 Implementation Plan 44
9.5 Testing and Validation Plan 44
9.6 Deployment Plan 45
9.7 Tentative Timeline 45
9.8 Expected Final Deliverables 45

10 Conclusion **47**

A Proposal Action Register **49**

B Detailed Requirements **53**

C Detailed Test Cases **55**

D Repository and Contribution Evidence **57**

E Additional Design Diagrams **59**

F Meeting and Progress Evidence **65**

Bibliography **67**

List of Figures

4.1 Use case diagram of the AI-based traffic control system. 15

5.1 Implementation architecture and current status (green: complete; orange: Iteration 2). 21
5.2 Domain model (class diagram) of the traffic-management system. 24

6.1 SUMO simulation toolkit: intersection scenario running with the baseline controller. 28
6.2 YOLO26 ambulance detection running on road video. 29
6.3 EquiFlow live dashboard: simulated camera views, intersection signal state, ambulance priority panel and live metrics. 30
6.4 Project repository structure (vision, marl, sumo, comm, dashboard, docs). . 30

7.1 Confusion matrix of the best checkpoint on the validation split (rows: predicted class, columns: true class). 35
7.2 Training curves for run `cls_v4-2`: training loss, validation top-1 accuracy, validation loss, and top-5 accuracy (constant at 1.0 for two classes and not informative). 36
7.3 Sample validation predictions on cropped vehicle images from run `cls_v4-2`. 36

D.1 Commit history of the EquiFlow repository. 57

E.1 Controller state machine for emergency handling. 59
E.2 System architecture: modules, data flows, and implementation status (green: complete in Iteration 1; orange: planned for Iteration 2; blue: external input). 60
E.3 Activity diagram: normal operation, detection, safe preemption, passage, recovery, and return to normal. 61
E.4 Sequence diagram: an ambulance approaching a sequence of two intersections. 62
E.5 Data-flow diagram: images, model weights, simulation state, controller actions, and metric outputs. 63

F.1 Signed FYP meeting log (F26-095-EquiFlow), supervised by Dr. Suleman Khan. 66

List of Tables

1.1 Project modules, responsibilities and status . . . 3
1.2 Stakeholders and user classes . . . 4

2.1 Comparison of existing research lines and the proposed project . . . 8
2.2 Research findings and resulting design implications . . . 10

3.1 Proposal action register . . . 11
3.2 Work breakdown and responsibilities . . . 13
3.3 Progress against plan . . . 13

4.1 Detailed use cases . . . 16
4.2 Functional requirements: perception and traffic-state module . . . 16
4.3 Functional requirements: dynamic control and recovery module . . . 17
4.4 Non-functional requirements . . . 17
4.5 Acceptance criteria . . . 19
4.6 Requirements traceability . . . 19

5.1 Architectural components, inputs, outputs and status . . . 22
5.2 Architecture decisions and rationale . . . 23
5.3 Dataset composition (images per class and split) . . . 25

6.1 Implementation evidence summary (Iteration 1) . . . 28
6.2 Repository modules . . . 31
6.3 Tools and technologies . . . 32

7.1 Test cases and current status . . . 34
7.2 Validation-split confusion matrix (best checkpoint) . . . 34
7.3 Validation-split metrics derived from the confusion matrix . . . 35

8.1 Individual responsibilities . . . 40
8.2 Milestones and evidence . . . 40
8.3 Risks and mitigation . . . 41

9.1 Remaining deliverables by module . . . 43
9.2 Remaining iterations . . . 44
9.3 Tentative timeline for FYP-2 . . . 45

A.1 Detailed proposal action register 49
A.2 Dataset sources and licences 50
A.3 Supervisor feedback and team actions 51

B.1 Detailed requirements 53

C.1 Detailed test cases 55

D.1 Contribution of each team member 58

F.1 Supervisory meeting log (as recorded on the signed sheet) 65

Students' Submission

We submit this report as the FYP-1 mid-report for the project titled "**Smart Traffic and Surveillance Management System**". The report describes the problem, research basis, proposed system, requirements, design, the implementation completed so far, early validation evidence, and the remaining work for FYP-2.

Student 1: Abdul Hadi Ayaz Abbasi (23I-0707) Signature: 

Student 2: Muhammad Adeel (23I-0739) Signature: 

Student 3: Marvan Shahid (23I-0800) Signature: 

Anti-Plagiarism Statement

The students certify that this report is their own work and that all sources used in preparing the report have been appropriately acknowledged and referenced. External research, software, datasets, frameworks, and other contributions are acknowledged where applicable.

Authors' Declaration

We declare that the work documented in this report represents the work completed by the project team up to the FYP-1 mid-report stage. Contributions, tools, external components, and relevant sources are disclosed where applicable. The report distinguishes completed implementation (sensing, simulation toolkit, dashboard, repository) from work that is still planned (tracking, route-aware preemption, MARL control, and system integration). Performance figures are reported only where they were measured from a logged run.

Abstract

Urban traffic congestion delays emergency response and creates uneven waiting times across competing traffic movements. Fixed-time signals cannot adapt to changing demand, while emergency-vehicle preemption can shorten ambulance travel time at the cost of extra delay for other road users. This project develops an AI-based traffic-management system that combines camera-based ambulance detection, emergency signal preemption, adaptive signal control, and fairness-aware recovery after an ambulance passes. The work is motivated by a gap between emergency-preemption research, which emphasizes ambulance travel time, and fairness-aware signal-control research, which generally does not model emergency vehicles.

At the FYP-1 mid-report stage the sensing and simulation foundations are complete. A SUMO toolkit covers 8 intersection types, 5 vehicle classes and 4 demand scenarios, with fixed-time and actuated baseline controllers. A YOLO26-based ambulance recognition model has been trained on a labelled vehicle-image dataset of 32,632 images; on the 3,188-image validation split it reaches 98.2% accuracy, 95.6% ambulance precision and 91.0% ambulance recall (F1 = 0.932). A live dashboard displays ambulance detections on video, and the repository is organized into vision, marl, sumo, comm, dashboard and docs modules. ByteTrack tracking, route-aware preemption, fairness-aware multi-agent reinforcement learning (MARL), and FastAPI/MQTT integration are scheduled for Iteration 2 in FYP-2. Test-set detection results and traffic-control performance claims are deferred until measured.

Acknowledgments

The project team acknowledges the guidance and support received from our supervisor Dr. Suleman Khan, co-supervisor Dr. Rizwan, faculty members, and peers throughout the FYP process.

Executive Summary

The project addresses two connected traffic-control problems: giving ambulances timely passage, and preventing the resulting signal disruption from imposing excessive or persistently unequal delay on other traffic. A rule that simply turns an empty green approach red can help locally, but it does not recognize emergency vehicles, coordinate multiple intersections, or guarantee fair recovery after preemption. The project therefore treats detection, signal control, and post-preemption recovery as parts of one system.

The literature review splits related work into two groups. Emergency-vehicle methods study detection, routing, preemption, or green corridors and commonly report ambulance travel time and aggregate traffic effects. Fairness-aware signal-control methods study delay disparity, queue allocation, or multi-intersection coordination but generally omit emergency vehicles. Prior work also indicates that preemption can increase cross-traffic delay. This motivates an integrated evaluation that reports both emergency response and the cost imposed on ordinary road users.

The proposed architecture has four stages (Figure 5.1): (1) sensing, (2) simulation, (3) tracking and control, and (4) integration. Stages 1 and 2 are complete. The sensing stage contains a YOLO26 ambulance recognition model and a live dashboard video view. The simulation stage contains a SUMO toolkit with 8 intersection types, 5 vehicle classes, 4 demand scenarios, and fixed-time and actuated baselines. Stages 3 and 4, which cover ByteTrack tracking, route-aware preemption, fairness-aware MARL, and FastAPI/MQTT dashboard integration, are planned for Iteration 2.

On the validation split, the trained classifier correctly labels 3,131 of 3,188 images (98.2%) against a majority-class baseline of 86.4%. It detects 394 of 433 ambulances (recall 91.0%) with 18 false alarms among 2,755 other-vehicle images (precision 95.6%). Because missed ambulances are the costlier error, improving recall and evaluating on the held-out test split are the first detector tasks in FYP-2. FYP-2 will then focus on tracking, preemption and recovery, MARL control, integration, repeated SUMO experiments against the baselines, sensitivity analysis, and final documentation.

List of Abbreviations

AI	Artificial Intelligence
CV	Computer Vision
EMV	Emergency Motor Vehicle / Emergency Vehicle
FN, FP, TN, TP	False negative, false positive, true negative, true positive
FYP	Final Year Project
FR	Functional Requirement
MAPPO	Multi-Agent Proximal Policy Optimization
MARL	Multi-Agent Reinforcement Learning
MQTT	Message Queuing Telemetry Transport
NFR	Non-Functional Requirement
RL	Reinforcement Learning
SUMO	Simulation of Urban MObility
TSC	Traffic Signal Control
YOLO	You Only Look Once object-detection model family

Chapter 1

Introduction and Project Vision

1.1 Project Overview

The project builds a traffic-management system that adapts signal phases to observed traffic conditions and grants controlled priority to ambulances. When an ambulance is detected approaching a signalized intersection, the system assesses its approach and the current signal state, safely transitions to a compatible phase, and coordinates a priority path through selected intersections where the network and simulation permit. After passage, the controller manages the queues and waiting times created by the interruption instead of simply returning to a fixed cycle.

At mid-report stage the project has a working sensing stage (ambulance recognition and a live detection dashboard), a working simulation stage (SUMO toolkit with baseline controllers), and a modular repository. Tracking, route-aware preemption, fairness-aware MARL control, and system integration are the next iteration (Chapter 6).

1.2 Problem Statement

Fixed-time traffic signals allocate green time according to predefined schedules and may waste green time on an empty approach while queues build elsewhere. Emergency preemption can shorten ambulance delay, but it interrupts ordinary traffic and may create secondary congestion or disproportionate waiting for cross traffic. Existing research often optimizes emergency travel time or ordinary-traffic fairness separately. The project investigates how ambulance detection and priority can be integrated with adaptive, fairness-aware signal control so that emergency passage is expedited while ordinary traffic receives timely and equitable recovery.

1.3 Problem Context and Motivation

In busy urban networks, demand varies by time, direction, road geometry, and incident conditions. Emergency vehicles must navigate this demand under time pressure. In Pakistan, mixed traffic—cars, buses, motorcycles, rickshaws, and vehicles with locally specific ambulance

markings—adds perception and control challenges. A practical prototype should therefore assess detection uncertainty, varying demand, queue spillback, and the impact of preemption on all approaches. Simulation offers a safe, repeatable way to compare policies before any real-world deployment.

1.4 Project Vision

To develop and evaluate a prototype that detects approaching ambulances, allocates safe signal priority, adapts ordinary signal timing to traffic demand, and restores fair service after preemption, using measurable performance criteria and reproducible simulation.

1.5 Project Objectives

- Review and compare research on ambulance detection, emergency-vehicle preemption, adaptive traffic-signal control, and fairness-aware control. (**Done**)
- Prepare or select a labelled dataset for ambulance recognition and document its classes and splits. (**Done**; sources and licences are documented in Section 5.7 and Appendix A, action A3)
- Develop a perception pipeline that identifies ambulances and, with ByteTrack, tracks them across frames. (recognition **Done**; tracking **Iteration 2**)
- Model signalized road networks and traffic demand in SUMO, including emergency-vehicle trips and ordinary flows. (**Done**)
- Implement baseline signal policies and an emergency-priority mechanism with safe phase transitions. (baselines **Done**; preemption **Iteration 2**)
- Design and evaluate a fairness-aware recovery policy that limits excessive waiting after preemption. (**Iteration 2**)
- Compare the proposed policy with fixed-time and adaptive baselines using emergency and non-emergency metrics. (**Iteration 2**)
- Document limitations, reproducibility details, and conditions required before real-world deployment. (ongoing)

1.6 Project Scope

Included in scope

- A research prototype for camera/video-based ambulance recognition and tracking.
- Dynamic signal timing based on traffic state, including shortening an unused green phase when safe.

- Emergency priority/preemption and a simulated green corridor across selected intersections.
- Fairness-aware service recovery for approaches delayed by preemption, with monitoring of maximum and high-percentile waiting time.
- SUMO-based simulation and quantitative comparison with baseline controllers.
- A dashboard demonstration and a report documenting assumptions, tests, and limitations.

Outside the initial scope

- Direct control of live public-road traffic lights or connection to municipal signal infrastructure.
- A guarantee of real-world safety based solely on simulation results.
- Full city-scale deployment, production-grade traffic-camera integration, or ambulance hardware installation.
- Recognition of every emergency-vehicle type beyond the defined class specification.

1.7 Project Modules

Table 1.1: Project modules, responsibilities and status

Module	Responsibility	Key output	Status
M1 – Data and perception	Dataset preparation, YOLO26 ambulance recognition, ByteTrack tracking.	Ambulance detections with confidence; tracks.	Recognition **Done**; tracking **Iteration 2**
M2 – Traffic-state estimation	Lane queues, occupancy, waiting time, signal phase, approaching emergency vehicles.	Structured state for control.	**Iteration 2**
M3 – Dynamic signal control	Allocate green time; skip empty approaches; adapt phase duration.	Phase and duration decisions.	Baselines **Done**
M4 – Emergency priority	Validate detections, select compatible phase, route-aware preemption.	Preemption request and priority route.	**Iteration 2**
M5 – Fair recovery / MARL	Restore service after preemption; balance delay; fairness-aware MARL.	Recovery policy.	**Iteration 2**
M6 – Simulation and evaluation	SUMO scenarios, baselines, metrics, repeated runs.	Scenarios **Done**; results pending.	Toolkit **Done**

1.8 Stakeholders and User Classes

Table 1.2: Stakeholders and user classes

Stakeholder	Interest / responsibility
Emergency services and ambulance drivers	Reduced emergency travel time and predictable passage.
Ordinary road users	Reasonable delay, safe transitions, and no approach starved indefinitely.
Traffic-control authorities	Reliable, auditable control logic and safe operational constraints.
Project team	Implement, test, document, and maintain the prototype.
Supervisor and evaluators	Assess technical depth, evidence, scope, and research contribution.
Researchers / future developers	Reuse documented methods, assumptions, and reproducible evaluation.

1.9 Success Criteria

These are the evaluation criteria for the project; only those marked as achieved are reported as results.

- The detector is evaluated on a held-out test set with precision, recall, and F1 (and mAP for the detection stage), including false-positive and false-negative analysis. (validation split done; test split **Iteration 2**).
- The simulation runs repeatably for fixed seeds, traffic demand, and vehicle routes.
- The emergency-priority policy reduces ambulance travel time relative to the fixed-time baseline in defined scenarios.
- Ordinary-traffic impact is reported: mean delay, queue length, throughput, and waiting-time distribution.
- The recovery policy is compared with a no-recovery/naive-return baseline for maximum/high-percentile wait and fairness.
- All reported experiments include configuration, scenario, number of runs/seeds, and metric definitions.
- No live signal-control claim is made; safety and deployment limitations are documented.

1.10 Constraints and Assumptions

- The project is an academic prototype with limited time, compute, and access to traffic infrastructure.
- The detector operates on small cropped vehicle images (96×96 pixels in the current run), so fine visual detail is limited.

- The dataset is class-imbalanced (about 17.5% ambulances), and its class balance differs between splits.
- SUMO traffic behavior depends on network quality, routes, demand, and driver-model assumptions.
- The controller must obey phase-conflict and clearance-time constraints in simulation.
- Camera visibility, occlusion, lighting, weather, and local vehicle appearance affect detection reliability.

1.11 Ethical, Legal and DEI Considerations

The prototype must prioritize safety and must not be presented as suitable for live deployment without engineering validation and authorization. If real traffic-camera footage is used, privacy, access rights, retention, and data-protection requirements will be assessed. Dataset licenses and source attribution must be respected. Evaluation examines whether particular approaches or traffic flows experience excessive waiting, instead of optimizing only the emergency vehicle. The dashboard and documentation make uncertainty and failure conditions visible.

Chapter 2

Research and Existing Solutions

2.1 Research Method / Approach

A literature review and gap analysis was used as the research baseline. The review separates studies into fairness-aware traffic-signal control, emergency-vehicle preemption/green corridors, and ambulance detection. For each study, the analysis records whether it addresses fairness, emergency priority, multi-intersection networks, perception-based detection, and post-preemption recovery or waiting-time bounds. Studies available only at abstract level are treated cautiously and must be verified from the full text before the final defense.

2.2 Existing Solutions / Products

Research methods fall into three main categories. First, fixed-time and rule-based adaptive controllers use predefined phases or traffic-pressure/queue rules. Second, reinforcement-learning methods optimize phase decisions from traffic state, sometimes coordinating multiple intersections and incorporating fairness [1, 2, 4, 10]. Third, emergency systems detect or receive the position of an ambulance and preempt signals or construct a green corridor [7, 8]. Commercial priority systems also exist, but vendor claims require independent verification and are not used as evidence in this report.

2.3 Comparison of Existing Solutions

Table 2.1: Comparison of existing research lines and the proposed project

Research line / example	Strength	Limitation relevant to this project
Fixed-time signal control	Simple, predictable, easy baseline.	Cannot respond to rapidly changing demand or empty approaches.
Raeis & Leon-Garcia – fairness-aware DRL [4]	Defines delay/throughput fairness; reports tail waiting metrics in SUMO.	Single-intersection focus; no emergency vehicles; fairness is a reward penalty, not a hard guarantee.
FAirLight – constrained RL [2]	Treats fairness and environmental performance as constraints.	Reviewed abstract does not combine the constraints with emergency preemption.
FairSignal / FA-MAPPO [1, 10]	Multi-intersection or multi-agent fairness-aware control.	Emergency priority and post-preemption recovery not established in the reviewed summaries.
EMVLight [7]	Couples emergency-vehicle routing and signal preemption across a network in SUMO.	Focuses on emergency travel time and efficiency; does not optimize fairness among ordinary drivers.
RTAIAED ambulance detector [8]	Visual/audio cues with temporal confirmation.	Small, localized dataset; local ambulance markings need validation.
YOLO-based detection systems [5]	Practical, real-time visual pipeline.	Detection alone gives no network-wide fair control or recovery.
Proposed project	Combines ambulance detection and tracking, preemption, adaptive control, and fairness-aware recovery in one evaluation.	Needs careful dataset handling, simulation design, and robust integration.

2.4 Limitations and Gaps

The central gap is the limited integration of emergency priority with explicit fairness for ordinary traffic. Fairness studies generally omit emergency vehicles, while emergency-preemption studies commonly report emergency travel time and average traffic effects without optimizing individual waiting-time equity. Detection systems may trigger a green phase but do not necessarily model network-wide queue recovery. Some control studies assume the emergency vehicle's location is already known and do not evaluate perception errors. Local mixed traffic and Pakistani ambulance markings are also underrepresented in the reviewed studies.

2.5 Proposed Solution

The project proposes an integrated perception-and-control pipeline. A camera or video stream supplies candidate ambulance detections; low-confidence results are filtered and, with ByteTrack [11], confirmed across consecutive frames. The traffic-state layer maps the detected approach to a simulated intersection and maintains queue and wait estimates. An emergency-priority manager requests a compatible phase and prepares downstream intersections along the ambulance route. After passage, the controller resumes adaptive service and gives recovery priority to approaches with high accumulated waiting or queue pressure, using a fairness-aware MARL policy compared against simpler rule-based policies.

2.6 Complex Computing Problem

The problem is complex because it combines uncertain visual perception, dynamic and partially observed traffic states, competing objectives, safety constraints, and temporal coordination across intersections. A decision that benefits one vehicle may increase delays elsewhere; preemption effects can propagate through queues and cause spillback. The controller must balance emergency travel time, average network performance, and the tail of the waiting-time distribution while respecting valid signal phases and clearance intervals. With reinforcement learning, the action space, reward/constraint design, training stability, and evaluation under unseen demand add further complexity.

2.7 Technical Challenges

- Class imbalance and split shift in the ambulance dataset (Section 5.7), and preventing train/validation leakage between similar frames.
- Separating ambulances from visually similar vans, trucks, and non-emergency vehicles.
- Converting detections into a reliable approach direction and time-to-intersection estimate.
- Synchronizing detector output with SUMO traffic state and signal phases.
- Avoiding conflicting green phases and enforcing minimum green, yellow, and all-red clearance.
- Designing a fairness objective that does not undermine emergency response or cause unsafe oscillation.
- Comparing learning-based policies fairly with tuned baselines under repeated stochastic scenarios.
- Reporting tail delay and maximum wait, not only average travel time.

2.8 Research Findings and Design Decisions

Table 2.2: Research findings and resulting design implications

Finding from literature review	Design implication
Emergency preemption can increase cross-traffic delay.	Report ordinary-traffic effects and implement a recovery mechanism.
Fairness is often a soft reward term.	Evaluate reward-based fairness and a bounded-wait or constraint-inspired recovery rule.
Many detection systems use locally specific ambulance data.	Train and validate on images representative of the target setting (done for the current dataset).
Some control studies assume perfect emergency-vehicle location.	Document perception errors; test false-positive/false-negative scenarios.
Multi-agent control coordinates intersections but adds complexity.	Build and benchmark rule-based baselines first (done); add MARL in Iteration 2.
SUMO is widely used in relevant traffic-control research [6].	Use SUMO for repeatable network and demand experiments (done).

Chapter 3

Proposal Feedback and Iteration Plan

3.1 Proposal Feedback Summary

Supervisor feedback was received over five meetings with Dr. Suleman Khan (Table F.1). In the first meeting the team was asked to review papers related to the project idea; in the second, the project scope and modules were revised; in the third, dataset augmentation and preprocessing were discussed and a dashboard showing live updates was requested; and in the fifth, the supervisor asked for multiple real Pakistani traffic scenarios in the simulation. The meeting-wise feedback and the action taken are listed in Appendix A (Table A.3). The action register below (Table 3.1) lists the research- and scope-driven actions taken from the proposal.

3.2 Proposal Action Register

Table 3.1: Proposal action register

No.	Required action / issue	Action taken	Status / evidence
1	Define a focused contribution beyond generic adaptive traffic lights.	Focus on ambulance priority with fairness-aware recovery.	**Done**; scope in Chapter 1.
2	Justify the research gap with a targeted comparison.	Compared fairness-control, emergency-preemption, and detection literature.	**Done**; Table 2.1.
3	Do not claim detection without labelled data.	Trained and validated a labelled ambulance / other-vehicle model.	**Done**; Section 7.4. Dataset-licence notes pending.
4	Evaluate impact on ordinary traffic as well as ambulance travel time.	SUMO toolkit with fixed-time and actuated baselines built.	**Partial**; multi-metric experiments in **Iteration 2**.
5	Make recovery after preemption an explicit feature.	Recovery and fairness-aware MARL designed.	**Iteration 2**; implementation pending.

3.3 Iteration Strategy

We develop the system incrementally. Each iteration produces a runnable artifact and supporting evidence. Iteration 1 established the foundations that everything else depends on: sensing (ambulance recognition and a dashboard view) and simulation (SUMO scenarios and baselines). Iteration 2 adds tracking and control and integrates the modules. A learning-based controller is introduced only on top of the stable simulation state, action interface, and baseline metrics from Iteration 1.

3.4 Iteration 1 Plan

Objectives: build the sensing and simulation foundations and a modular repository.

- Train a YOLO26 ambulance recognition model on a labelled dataset. **Done**
- Display detections on video in a live dashboard. **Done**
- Build SUMO scenarios (8 intersection types, 5 vehicle classes, 4 demand scenarios). **Done**
- Implement fixed-time and actuated baseline controllers. **Done**
- Organize the repository as vision, marl, sumo, comm, dashboard, and docs. **Done**

Deliverables and evidence: trained model and training logs, dashboard demonstration, SUMO toolkit, repository (Chapter 6).

3.5 Iteration 2 Plan

Objectives: add tracking and control, then integrate the modules.

- Add ByteTrack tracking so detections persist across frames.
- Implement route-aware preemption with safe phase transitions and a green corridor where feasible.
- Implement and train a fairness-aware MARL controller and a recovery policy.
- Connect modules through FastAPI and MQTT and link the dashboard to the live system.
- Complete formal detector evaluation (test split, latency) and run repeatable baseline comparisons.

Deliverables: integrated prototype, experiment scripts, preliminary results, and documented failure cases.

3.6 Work Breakdown and Responsibilities

Table 3.2: Work breakdown and responsibilities

Workstream	Primary responsibility	Supporting role / evidence
Research and requirements	Literature review, requirements, citations.	Literature matrix and meeting notes.
Data and detection (vision)	Dataset, model training, detector evaluation, tracking.	Training logs, confusion matrix, dataset counts.
Simulation and control (sumo, marl)	SUMO networks, demand, baselines, emergency routes, MARL.	Configuration and reproducible runs.
Integration (comm, dashboard)	FastAPI/MQTT interfaces, dashboard, end-to-end tests.	Code, logs, demonstrations.
Documentation and release (docs)	Report, version control, test evidence, packaging.	Repository and release tag.

3.7 Progress Against Plan

Table 3.3: Progress against plan

Work item	Status	Evidence
Problem and research gap	**Done**	Literature review; Chapter 2.
Dataset preparation	**Done**	32,632 labelled images in train/val/test splits (Table 5.3).
YOLO26 ambulance recognition	**Done**	Training run `cls_v4-2`; Figures 7.1–7.3.
Live detection dashboard	**Done**	Figure 6.3.
SUMO toolkit and baselines	**Done**	Figure 6.1; fixed-time and actuated controllers.
Repository structure	**Done**	Figure 6.4.
ByteTrack tracking	**Iteration 2**	Planned.
Route-aware preemption	**Iteration 2**	Planned.
Fair recovery / MARL	**Iteration 2**	Planned.
FastAPI / MQTT integration	**Iteration 2**	Planned.
Test-split and latency evaluation	**Iteration 2**	Planned.

3.8 Delays, Deviations and Recovery Plan

Iteration 1 was scoped to the two stages that other stages depend on (sensing and simulation), so tracking, preemption, MARL, and integration were deliberately moved to Iteration 2 instead of being started in parallel. The formal detector evaluation (test split, latency) was also deferred; only validation-split metrics are available at this point. No dated project log

was available, so schedule slippage is not quantified here. The anticipated schedule risks are MARL training cost, ByteTrack integration with the recognition model, and integration effort across modules. The recovery strategy is to rely on the working rule-based baselines and SUMO evaluation, and to fall back to a transparent heuristic recovery policy if MARL training is too costly.

Chapter 4

Software Requirements

4.1 User Interactions / Use Cases

The prototype is an experimental system rather than a public-facing traffic-control product. The main user is the project operator/researcher who configures scenarios, starts experiments, and inspects results. In the dashboard, the operator can watch video with ambulance detections, inspect detected emergency vehicles, observe signal states, and view performance metrics.

4.2 Use Case Diagram

Figure 4.1 shows the use cases of the system. The operator configures a scenario, runs the simulation, inspects ambulance detection, observes priority/preemption and recovery, and exports metrics and logs. Two external inputs feed the system: the road network, demand files and controller configuration, and the video stream or simulated emergency-vehicle state.

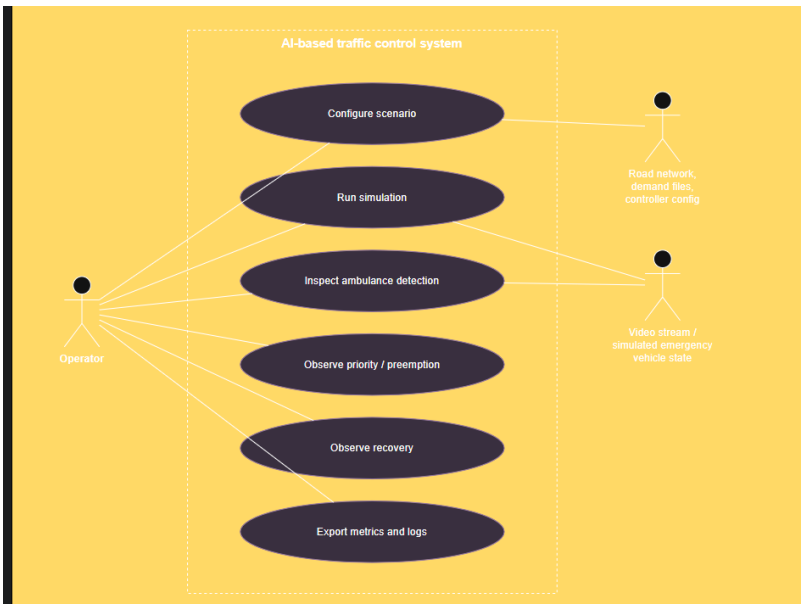


Figure 4.1: Use case diagram of the AI-based traffic control system.

4.3 Detailed Use Cases

Table 4.1: Detailed use cases

Use case	Precondition	Main flow	Postcondition
UC-01 Configure scenario	Network and configuration files exist.	Operator selects network, demand, emergency routes, seed, controller.	Scenario saved and reproducible.
UC-02 Detect ambulance	Video frame and trained model available.	Model predicts class and confidence; thresholds and temporal checks applied.	Detection event or no-event recorded.
UC-03 Request priority	Ambulance mapped to an approach; signal state known.	Priority manager checks phase compatibility and requests a safe transition.	Priority event and phase transition logged.
UC-04 Recover ordinary traffic	Ambulance has passed or preemption cleared.	Controller assesses queues and waits and restores service fairly.	Interrupted approaches served under the recovery policy.
UC-05 Evaluate run	Simulation has completed.	Metrics computed and compared with baselines.	Run summary, configuration, and results exported.

4.4 Functional Requirements

4.4.1 Perception and traffic-state module

Table 4.2: Functional requirements: perception and traffic-state module

ID	Requirement	Priority	Status
FR-01	The system shall accept supported image/video input or simulated detection events.	Must	**Done** (video)
FR-02	The perception component shall output ambulance class and confidence (and bounding-box coordinates in detection mode).	Must	**Partial**
FR-03	The system shall record detection confidence and timestamp for audit and evaluation.	Must	**Iteration 2**
FR-04	The traffic-state module shall track phase, lane/approach queue, waiting time, and emergency approach state.	Must	**Iteration 2**
FR-05	The system shall handle missing or low-confidence detections without issuing an unsafe phase change.	Must	**Iteration 2**

4.4.2 Dynamic control and recovery module

Table 4.3: Functional requirements: dynamic control and recovery module

ID	Requirement	Priority	Status
FR-06	The controller shall support configurable signal phases and minimum/maximum green durations.	Must	**Done** (baselines)
FR-07	The controller shall end an unused green early only when phase-transition and clearance rules permit.	Should	**Partial**
FR-08	The emergency manager shall request priority for a validated ambulance approaching an intersection.	Must	**Iteration 2**
FR-09	The system shall log preemption start, target approach, clearance, ambulance passage, and recovery completion.	Must	**Iteration 2**
FR-10	After preemption, the controller shall consider queue length and waiting time when allocating service.	Must	**Iteration 2**
FR-11	The evaluation shall compare proposed control against fixed-time and at least one adaptive baseline.	Must	**Partial** (baselines built)
FR-12	The system shall export metrics and configuration for repeatable analysis.	Must	**Iteration 2**

4.5 Non-Functional Requirements

Table 4.4: Non-functional requirements

Category	Requirement
Reliability	Control decisions deterministic for a fixed state and configuration unless the policy is stochastic; failures logged.
Safety	No conflicting movements green simultaneously; yellow/all-red and compatibility constraints enforced.
Usability	Documented configuration and run commands; results understandable without reading source code.
Performance	Perception/control loop measured against intended frame rate and simulation step; real-time claims need evidence.
Maintainability	Perception, state, control, recovery, and evaluation are separate modules (reflected in the repository layout).
Reproducibility	Record seeds, model weights/version, dataset split, simulation settings, and dependency versions.
Security and privacy	Do not store identifiable footage unnecessarily; control access to restricted data.
Portability	Workflow runs on the team's Windows/Linux environment with documented dependencies.

4.5.1 Reliability

The controller should fail safely in simulation if perception input is missing, stale, or invalid. Detection events must be timestamped and stale events must not request priority indefinitely.

4.5.2 Usability

The system should provide clear configuration instructions, consistent log messages, and a concise run summary of ambulance travel time, ordinary-traffic delay, queue metrics, and fairness measures. The dashboard already provides a visible detection view.

4.5.3 Performance

Performance must be measured empirically. We will record detector inference time, simulation throughput, controller decision latency, and the number of completed scenario runs. No real-time guarantee is claimed until measured on the target hardware. (The only timing recorded so far is the 30-epoch training run: about 20 minutes on one GPU.)

4.5.4 Security

Use trusted datasets and packages, pin software versions where practical, avoid embedding credentials in source code, and document external data sources and APIs. Real traffic-camera data must be handled under applicable privacy and access rules. Network interfaces added in Iteration 2 (FastAPI, MQTT) must be authenticated before any deployment beyond a local demonstration.

4.6 Acceptance Criteria

Table 4.5: Acceptance criteria

Requirement area	Acceptance check
Dataset	Class folders and IDs are consistent; splits documented and free of duplicate source images across train/val/test.
Detection	Precision, recall, F1 (mAP for detection stage) and FP/FN examples reported on a held-out test set.
Signal safety	Automated tests confirm no conflicting phases are green simultaneously and clearance intervals are respected.
Emergency priority	A scripted emergency trip triggers a logged priority event and completes under the defined policy.
Recovery	After preemption, ordinary service resumes and recovery interval and waiting-time distribution are recorded.
Evaluation	At least three repeatable runs per scenario/seed configuration or a justified statistical design; compared with baselines.
Reproducibility	A second team member can reproduce a documented run from configuration and instructions.

4.7 Requirements Traceability

Table 4.6: Requirements traceability

Requirement IDs	Design / module (repo)	Verification method
FR-01–FR-03	M1 Perception (vision)	Detector validation/test sets, logged detection examples.
FR-04–FR-05	M2 State estimation	Unit tests with known queue/phase inputs and stale/missing detections.
FR-06–FR-07	M3 Dynamic control (sumo)	Phase-transition tests; empty/occupied approach scenarios.
FR-08–FR-09	M4 Emergency priority	Scripted ambulance arrival; event-log inspection.
FR-10	M5 Fair recovery (marl)	Preemption scenarios comparing wait and recovery metrics.
FR-11–FR-12	M6 Evaluation (sumo, docs)	Reproducible benchmark runs and exported result files.

Chapter 5

System Design

5.1 System Architecture

The architecture is a four-stage pipeline with a feedback loop (Figure 5.1). In video-driven operation, frames are processed by the detector; detections are validated, tracked, and mapped to a road approach. In simulation, SUMO supplies ground-truth vehicle position and traffic state for control experiments, while a separate perception experiment measures detector performance. These two modes are not conflated: using SUMO's known emergency-vehicle state is useful for control debugging but does not validate camera-based detection.

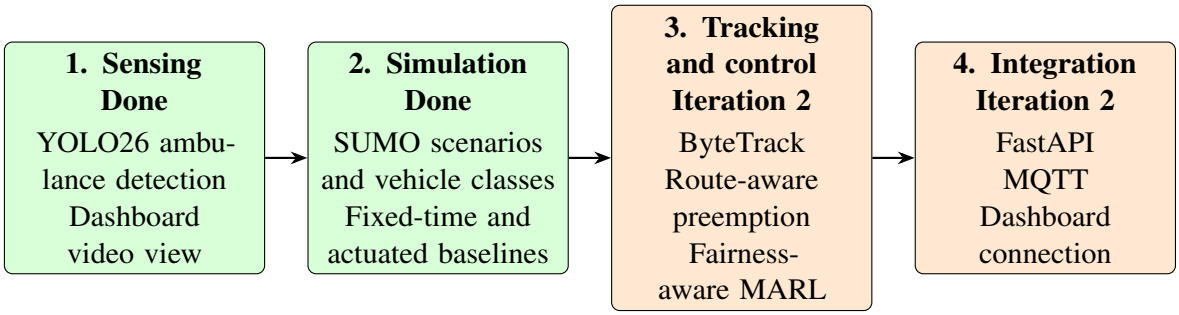


Figure 5.1: Implementation architecture and current status (green: complete; orange: Iteration 2).

5.2 Architectural Components

Perception and control are loosely coupled through a documented event interface (to be carried over FastAPI and MQTT). This enables separate testing: the detector is evaluated on a labelled image set, while the controller is first tested with reliable simulated emergency-vehicle positions. The integrated system will later introduce detection uncertainty and latency to measure robustness. The repository mirrors this separation (Table 6.2).

Table 5.1: Architectural components, inputs, outputs and status

Component	Inputs	Outputs	Status
Data / detector	Images/video, trained weights	Ambulance class, confidence, timestamps	**Done**
Tracker (ByteTrack)	Per-frame detections	Persistent track IDs	**Iteration 2**
State estimator	Vehicle/queue data, signal phase, detections	Traffic state; emergency approach estimate	**Iteration 2**
Priority manager	Emergency event, network map, current phase	Priority request, approach, preemption state	**Iteration 2**
Signal controller	Traffic state, phase constraints, policy	Next phase and duration	Baselines **Done**
Recovery manager (MARL)	Preemption completion, queue/wait state	Recovery service priorities	**Iteration 2**
SUMO scenarios	Network, routes, demand, phase commands	Simulation state, trajectories	**Done**
Comm layer (FastAPI, MQTT)	Detections, state, commands	Messages between modules	**Iteration 2**
Dashboard	Video, detections	Live detection view	**Done**
Evaluation / logger	Events, trips, signals, configuration	CSV/JSON logs, metrics, plots	**Iteration 2**

- FastAPI and MQTT for inter-module communication and the dashboard connection (Iteration 2).

- CSV/JSON structured logs for experiment configuration and metrics.

5.3 Interfaces and Dependencies

- Python for orchestration, data preparation, experiment scripts, and evaluation.

- Ultralytics YOLO26 [5] for ambulance recognition; ByteTrack [11] for multi-object tracking.

- SUMO [6] for simulation; a SUMO control interface (e.g. TraCI) for signal control in Iteration 2.

5.4 Architecture Decisions and Rationale

Table 5.2: Architecture decisions and rationale

Decision	Rationale	Trade-off
Use SUMO for control evaluation	Safe, repeatable, configurable.	Realism limited by network and demand assumptions.
Build rule-based baselines before MARL	Simpler to validate and explain; gives a benchmark.	May not be globally optimal.
Separate detector from controller	Independent evaluation and easy replacement.	Needs an explicit interface and timestamp handling.
Binary classes: ambulance / `other_vehicle`	Matches the labelled dataset; focuses on the emergency class.	Does not tell whether the ambulance is responding to an emergency.
Lightweight `yolo26n-cls` at 96×96	Fast training (about 20 min) and likely real-time friendly.	Limited detail; larger input or model may raise recall.
Track tail-wait and fairness metrics	Average delay can hide severely delayed approaches.	Needs careful definitions and instrumentation.

5.5 Design Evolution

Since the proposal, the design has gained two concrete elements: ByteTrack tracking between detection and preemption, and an explicit integration layer (FastAPI, MQTT, dashboard). Recovery after preemption remains a first-class module, and the repository was organized into vision, marl, sumo, comm, dashboard and docs so that a simple controller can be replaced later by a learning-based one.

5.6 Domain Model

Core entities: Intersection, Approach/Lane, SignalPhase, Vehicle, EmergencyVehicle, DetectionEvent, Track, PreemptionEvent, ControllerDecision, SimulationScenario, and EvaluationRun. A DetectionEvent references a vehicle hypothesis, timestamp, confidence, and image coordinates. A PreemptionEvent records the affected intersection/approach, start and end times, and phases interrupted. An EvaluationRun records configuration, seed, policy version, model version, and output metrics. The entities and their relationships are shown in Figure 5.2.

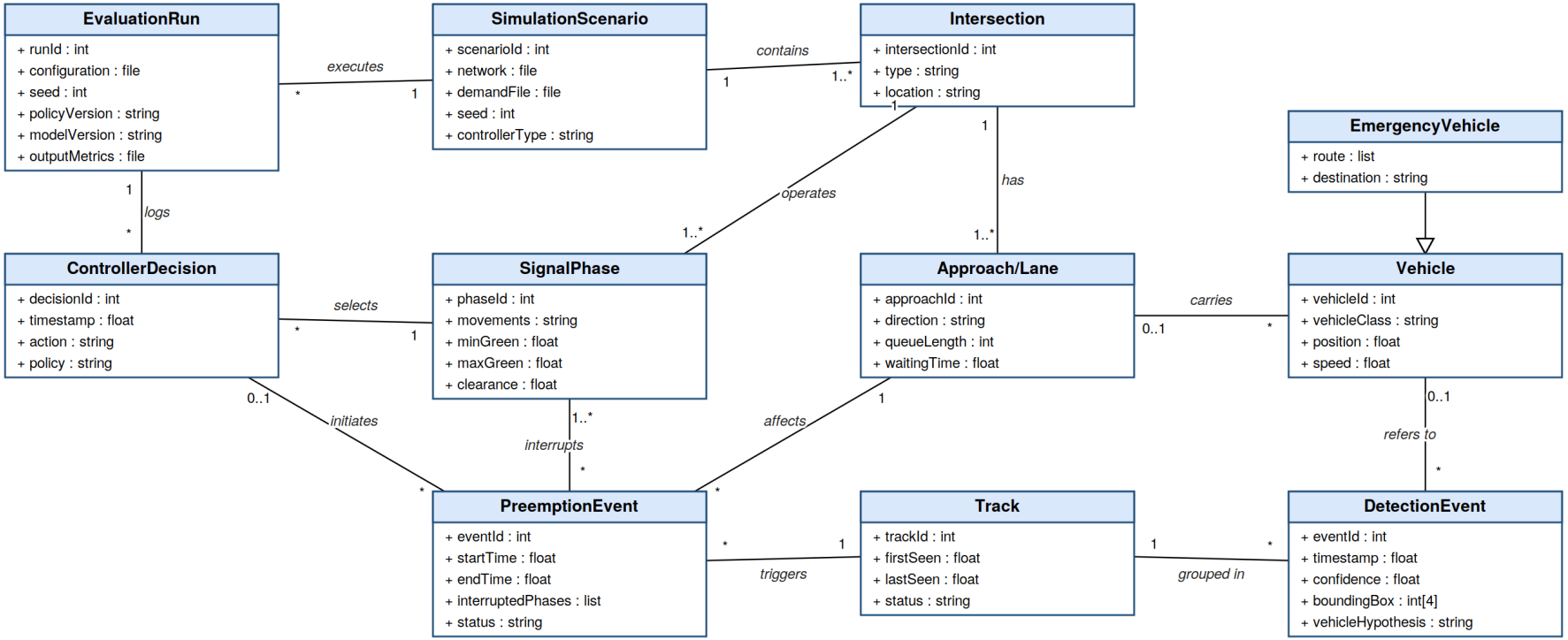


Figure 5.2: Domain model (class diagram) of the traffic-management system.

5.7 Data Design

The classification dataset is organized one folder per class (`ambulance`, `other_vehicle`) inside train, val and test splits. Table 5.3 gives the composition.

Table 5.3: Dataset composition (images per class and split)

Split	Ambulance	Other vehicle	Total	Ambulance %	Other : Amb.
Train	4,506	21,217	25,723	17.5%	4.7 : 1
Validation	433	2,755	3,188	13.6%	6.4 : 1
Test	768	2,953	3,721	20.6%	3.8 : 1
Total	**5,707**	**26,925**	**32,632**	**17.5%**	**4.7 : 1**

The dataset was assembled from several sources rather than a single benchmark, and part of it consists of vehicle crops extracted from larger images and from video frames, each saved as a separate image in its class folder. This gives more varied scenes, camera angles and vehicle appearances, but image quality, resolution and labelling quality differ between sources, and crops taken from the same video can look very similar. The name, link, licence and image count of each source are listed in Appendix A (Table A.2). The train, validation and test splits were made by random assignment of individual images and crops, not per source image or video. Crops taken from the same video or scene can therefore fall in more than one split, which may make the validation results optimistic, and the dataset acceptance criterion (Table 4.5) is not yet fully met. A source-level split will be made in FYP-2, before test-split results are reported.

Two properties of this data matter for evaluation. First, the dataset is imbalanced: a model that always answers "other vehicle" would score 82.5%, 86.4% and 79.4% accuracy on train, validation and test, so accuracy alone is misleading and precision, recall, and F1 for the ambulance class are reported. Second, the ambulance share differs across splits (17.5%, 13.6%, 20.6%), so precision is not directly comparable between validation and test, and a decision threshold tuned on validation may not transfer to test. The validation split also contains only 433 ambulances, so its recall estimate is noisy (a few images move it by several tenths of a percent).

For simulation, scenario configuration and per-run logs are kept separately. Recommended records include timestamp, simulation time, intersection ID, active phase, queue length per approach, waiting time, emergency-vehicle ID/position, preemption state, and controller action. Derived metrics are computed from raw logs with versioned scripts.

5.8 Process / Behavior Design

1. Read the current traffic state and signal phase.
2. Process the latest frame; validate candidate ambulance detections using confidence and temporal consistency (ByteTrack).

3. Estimate the ambulance approach and decide whether it is close enough to request priority.
4. Check the requested movement against conflict and clearance constraints.
5. Transition safely to the priority phase; coordinate downstream signals along the route where supported.
6. Monitor ambulance passage and clear the preemption state when it has passed.
7. Enter recovery mode: update queues and accumulated waits and select the next compatible phase using the recovery policy.
8. Return to ordinary adaptive control once recovery conditions are met; log all state changes and metrics.

The controller state machine for emergency handling is shown in Figure E.1 (Appendix E).

Chapter 6

Iteration 1: Design and Implementation

6.1 Iteration Objectives

Iteration 1 establishes the foundation on which preemption, tracking, and MARL are built: a trained ambulance recognition model, a visible detection interface, a SUMO toolkit with baseline controllers, and a modular repository. The status of each stage is summarized in Figure 5.1 and Table 6.1.

6.2 Selected Use Cases / Features

- UC-02 Detect ambulance: train the recognition model and show detections on video.
- UC-01 Configure scenario: build SUMO scenarios across intersection types, vehicle classes, and demand levels.
- Baseline control: fixed-time and actuated controllers for later comparison.
- Repository foundation for later integration.

6.3 Design

The sensing stage is a classification/recognition model trained on cropped vehicle images with two classes (ambulance, other vehicle) and exposed to the dashboard for live display. The simulation stage is a parameterized SUMO toolkit in which intersection type, vehicle mix, and demand are configurable so that policies can be compared under identical scenarios. Phase changes are governed by a signal-phase state machine instead of unconstrained switching.

6.4 Implementation

Four pieces of implementation evidence were produced in this iteration (Table 6.1).

Table 6.1: Implementation evidence summary (Iteration 1)

No.	Component	What is implemented	Status
#1	SUMO simulation toolkit	8 intersection types; 5 vehicle classes; 4 demand scenarios; fixed-time and actuated baseline controllers.	**Done**
#2	YOLO26 ambulance detection	Model trained on a labelled dataset; part of the vision module. Formal test-split precision/recall and latency evaluation is the next step.	**Done**
#3	Live detection dashboard	Dashboard displays ambulance detections on video; connects the trained model to a visible interface.	**Done**
#4	Project repository	Organized into vision, marl, sumo, comm, dashboard, docs; provides the modular base for integration.	**Done**

6.4.1 Implementation #1 – SUMO simulation toolkit

The toolkit provides scenario setup covering eight intersection types, five vehicle classes, and four demand scenarios, together with fixed-time and actuated baseline controllers. These form the reference against which emergency-priority and fairness-aware control will be measured.

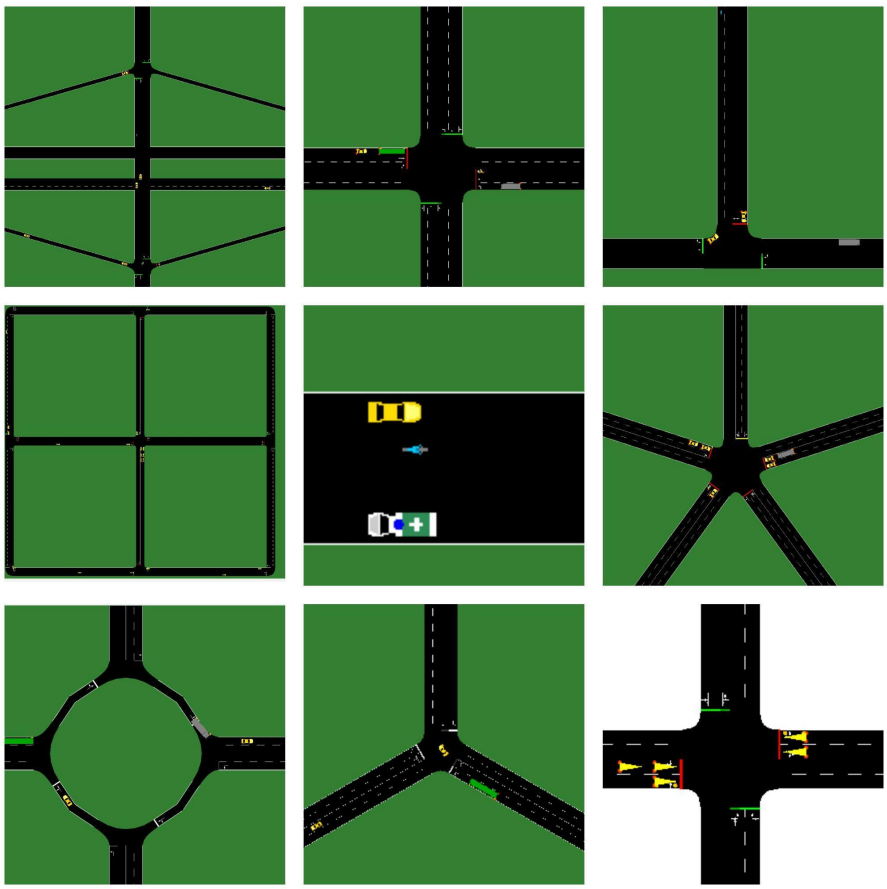


Figure 6.1: SUMO simulation toolkit: intersection scenario running with the baseline controller.

6.4.2 Implementation #2 – YOLO26 ambulance detection

The ambulance recognition model was trained with Ultralytics YOLO26 on the dataset in Table 5.3, which combines several sources and includes vehicle crops extracted from larger images and from video frames. The logged training run (`cls_v4-2`) used the pretrained `yolo26n-cls` model, 30 epochs, batch size 128, input size 96×96, seed 0, automatic optimizer selection (initial learning rate 0.01), and augmentation (horizontal flip 0.5, HSV jitter, RandAugment, random erasing 0.4). Training took about 20 minutes (1,210 s) on one GPU. Results are in Section 7.4.



Figure 6.2: YOLO26 ambulance detection running on road video.

6.4.3 Implementation #3 – Live ambulance detection dashboard

The EquiFlow dashboard (Figure 6.3) connects the trained model and the simulation to a visible interface. It shows a simulated camera view for each approach, the signal state of the main intersection (North and East, with the number of vehicles waiting and the waiting time), a panel announcing ambulance priority and the approach concerned (with the track ID and detection confidence), and a live-metrics panel (ambulances detected, priority runs, detection-to-green time, vehicles in view, and longest wait). It is the clearest user-facing evidence of current progress. The values in the screenshot come from simulated cameras and are illustrative of the interface; they are not evaluation results.

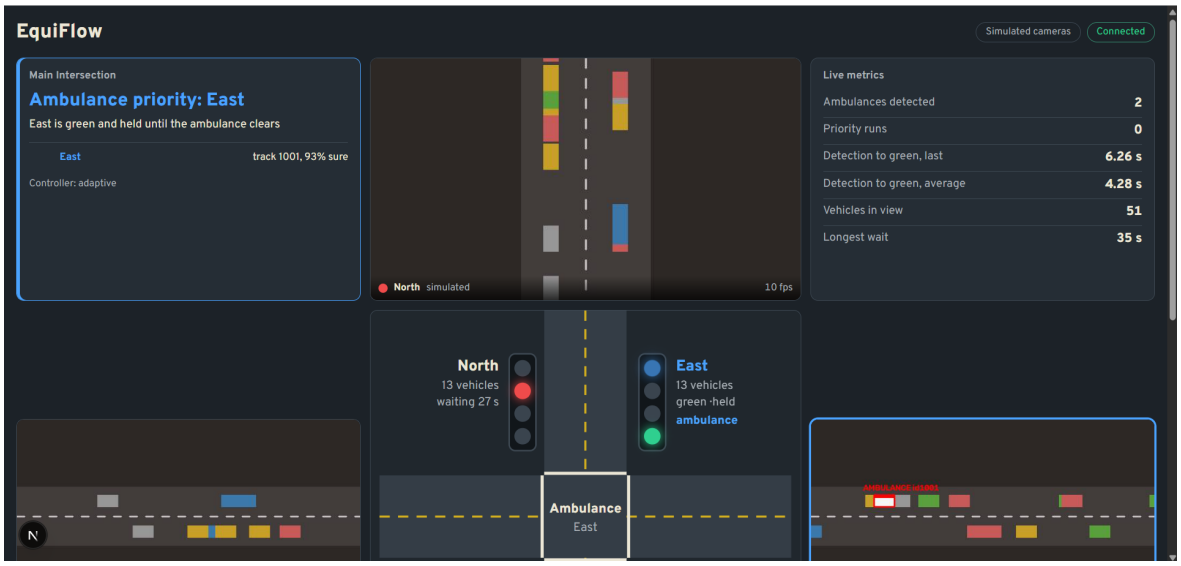


Figure 6.3: EquiFlow live dashboard: simulated camera views, intersection signal state, ambulance priority panel and live metrics.

6.4.4 Implementation #4 – Repository organization

The repository is organized into six modules (Table 6.2), giving a modular foundation for integration. MARL, tracking, and communication integration are still planned or in progress.

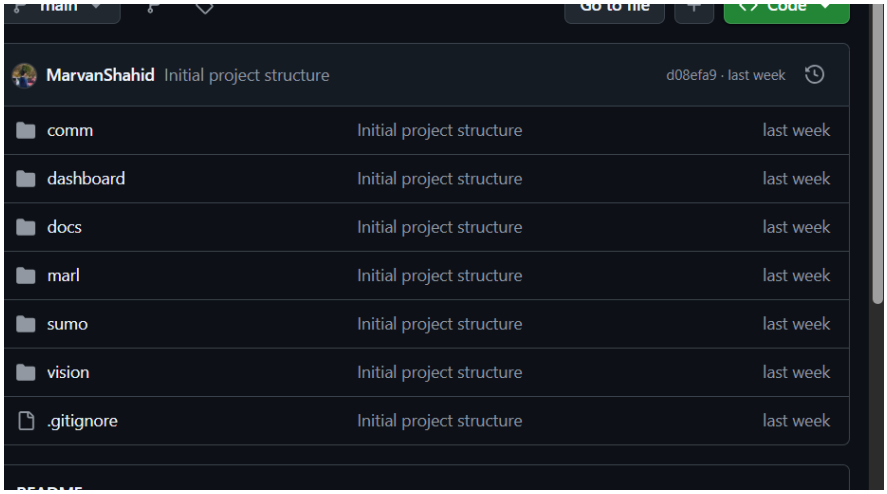


Figure 6.4: Project repository structure (vision, marl, sumo, comm, dashboard, docs).

6.5 Completed Features / Module

Completed: the SUMO toolkit with baselines, the trained ambulance recognition model, the live detection dashboard, and the repository structure. Not yet complete: ByteTrack tracking, route-aware preemption, fairness-aware MARL, communication integration, and formal detector evaluation on the test split.

Table 6.2: Repository modules

Directory	Purpose	Status
`vision`	Ambulance recognition model, training and inference; ByteTrack tracking later.	Recognition **Done**; tracking **Iteration 2**
`sumo`	Networks, scenarios, demand, and baseline controllers.	**Done**
`marl`	Fairness-aware multi-agent RL controller.	**Iteration 2**
`comm`	FastAPI and MQTT communication between modules.	**Iteration 2**
`dashboard`	Live detection dashboard.	**Done**
`docs`	Documentation and report material.	Ongoing

6.6 Technical Implementation Details

The vision module trains with the Ultralytics API, using class-per-folder data (train/val/test) and writing weights (`best.pt`, `last.pt`), per-epoch metrics (`results.csv`), and plots. Run settings are stored in `args.yaml` so a run can be reproduced. The sumo module holds the network, route, demand, and baseline signal configuration files. Detection events and phase commands will pass through the comm module in Iteration 2.

6.7 Algorithms and Technical Decisions

Recognition uses a lightweight pretrained convolutional classifier fine-tuned on the ambulance / other-vehicle data. Because of class imbalance, accuracy is not used as the headline metric. For control, the initial baselines are the fixed-time cycle and an actuated (demand-responsive) controller. Emergency preemption will override ordinary optimization only when a validated, tracked emergency event meets configured criteria, and recovery will prevent starvation through a waiting-time threshold or constrained objective. MARL is introduced only on top of these working baselines.

6.8 Tools, Frameworks and Technologies

Table 6.3: Tools and technologies

Tool / technology	Use	Status
Python	Orchestration, training, controller, evaluation.	In use
Ultralytics YOLO26	Ambulance recognition.	In use
SUMO	Traffic network and signal simulation.	In use
Dashboard	Live detection view.	In use
Git repository	Version control and modular layout.	In use
ByteTrack	Multi-object tracking.	**Iteration 2**
FastAPI, MQTT	Module communication and dashboard link.	**Iteration 2**
MARL framework	Fairness-aware control learning.	**Iteration 2**

6.9 Demonstration Evidence

Demonstration evidence in this report consists of the training outputs of run `cls_v4-2` (Figures 7.2–7.3) and the screenshots of the SUMO toolkit, dashboard, and repository (Figures 6.1–6.4). All figures are outputs of the actual system.

6.10 Iteration Outcome

Iteration 1 delivered the sensing and simulation foundations and a modular repository. The validation-split results show a strong recognizer (98.2% accuracy) whose weak point is ambulance recall (91.0%).

Carried into Iteration 2: formal test-split evaluation and latency measurement, recall improvement, ByteTrack tracking, preemption, MARL, and integration.

Chapter 7

Testing and Early Validation

7.1 Testing Strategy

Testing progresses from unit tests to component, integration, scenario, and comparative experiments. The same traffic demand and seeds are used across controller comparisons. For stochastic policies, multiple seeds are used and dispersion is reported with the means. Detector evaluation is kept separate from control evaluation.

7.2 Test Levels and Techniques

- Unit testing: label parsing, class-ID validation, phase compatibility, waiting-time and metric calculations.
- Component testing: detector on held-out images; controller with simulated state; SUMO adapter on small networks.
- Integration testing: detection or simulated ambulance event $\rightarrow$ preemption $\rightarrow$ passage $\rightarrow$ recovery.
- Scenario testing: no ambulance, one or several ambulances, low/high demand, empty approach, heavy queue, detection error.
- Performance testing: inference latency, simulation runtime, controller latency, repeatability.
- Comparative evaluation: fixed-time, actuated, emergency-only preemption, and recovery-aware control.

7.3 Test Cases

Table 7.1: Test cases and current status

ID	Scenario	Expected behavior	Status
TC-01	Image with visible ambulance and ground-truth label	Model output can be compared with the label.	**Partial** (validation split)
TC-02	Image without an ambulance	No ambulance event above threshold.	**Partial** (validation split)
TC-03	Low-confidence / one-frame detection	No unsafe immediate preemption; filtered or confirmed.	**Planned**
TC-04	Empty approach during green	Green shortened only if minimum-green/transition rules allow.	**Planned**
TC-05	Ambulance approaching red signal	Compatible priority phase requested; clearance observed.	**Planned**
TC-06	Ambulance passes intersection	Preemption ends; recovery begins.	**Planned**
TC-07	Approach with very high accumulated wait	Recovery gives service per the fairness rule.	**Planned**
TC-08	Conflicting phase request	Controller rejects or delays the unsafe change.	**Planned**
TC-09	No ambulance in scenario	Controller operates normally; baseline metrics recorded.	**Planned**
TC-10	Repeat run, same seed/configuration	Results reproducible within tolerance.	**Planned**

7.4 Test Results

Detector validation results. Table 7.2 and Figure 7.1 give the confusion matrix of the best checkpoint on the validation split (3,188 images; the counts 433 and 2,755 match the validation composition in Table 5.3). Table 7.3 lists the derived metrics.

Observations. Validation accuracy rose from 94.6% (epoch 1) to 98.2% (peak 98.21% at epoch 27; 98.18% at epoch 30). A difference of one image is not meaningful at this level.

Table 7.2: Validation-split confusion matrix (best checkpoint)

	True ambulance	True other vehicle
Predicted ambulance	394 (TP)	18 (FP)
Predicted other vehicle	39 (FN)	2,737 (TN)

Table 7.3: Validation-split metrics derived from the confusion matrix

Metric	Value	Calculation
Overall accuracy	98.2%	3,131 / 3,188
Majority-class baseline accuracy	86.4%	2,755 / 3,188
Ambulance precision	95.6%	394 / (394 + 18)
Ambulance recall	91.0%	394 / (394 + 39)
Ambulance F1-score	0.932	harmonic mean of precision and recall
Other-vehicle recall	99.35%	2,737 / 2,755

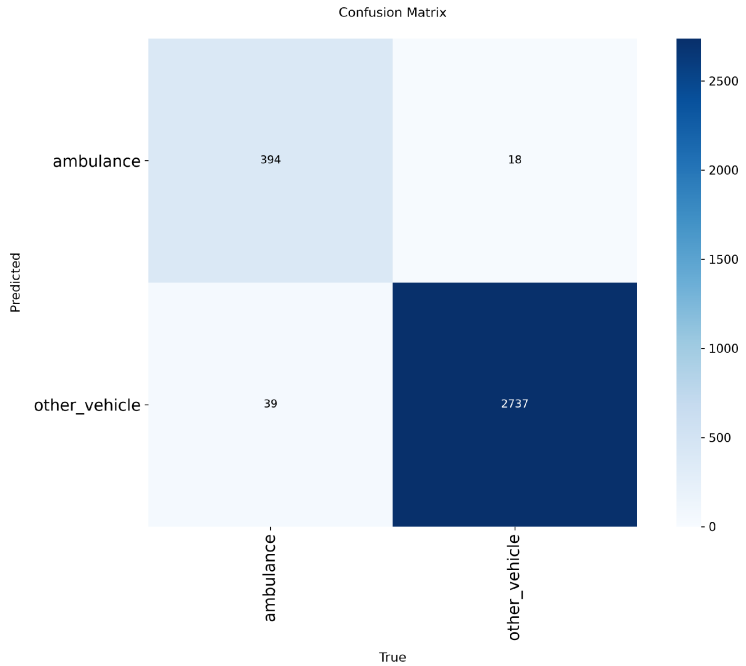


Figure 7.1: Confusion matrix of the best checkpoint on the validation split (rows: predicted class, columns: true class).

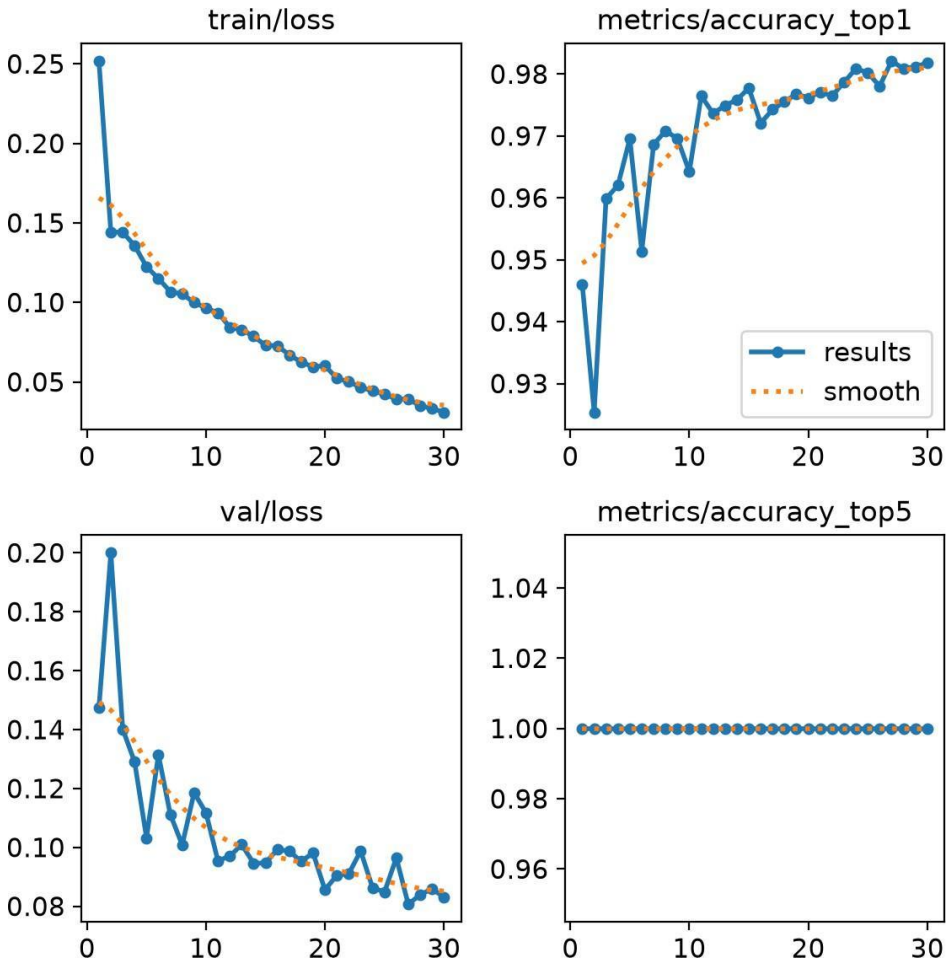


Figure 7.2: Training curves for run `cls_v4-2`: training loss, validation top-1 accuracy, validation loss, and top-5 accuracy (constant at 1.0 for two classes and not informative).



Figure 7.3: Sample validation predictions on cropped vehicle images from run `cls_v4-2`.

- Training loss fell steadily (0.252 to 0.031) while validation loss flattened around 0.08–0.10 after about epoch 20 (minimum 0.0807 at epoch 27), a mild sign of overfitting.
- Errors are asymmetric: 39 missed ambulances against 18 false alarms. Since a missed ambulance is the costlier error, recall is the weak spot, and lowering the ambulance decision threshold or increasing input size/model capacity are the first options to try.
- These are validation-split numbers. The best checkpoint was selected using this split, so they are slightly optimistic; the random, non-source-level split (Section 5.7) may inflate them further. Test-split results (768 ambulances, 2,953 other) are not yet reported.

Simulation and dashboard. The SUMO scenarios and baseline controllers run, and the dashboard displays detections on video. No traffic-performance numbers (delay, queue length, ambulance travel time, fairness) are reported yet because the comparison experiments are scheduled for Iteration 2.

7.5 Acceptance Testing / Validation

Validation includes visual inspection of predictions (Figure 7.3), confirmation of phase transitions, verification of ambulance travel-time measurement, and review of post-preemption waiting-time distributions. Learned policies will be evaluated on scenarios not used in training.

7.6 Edge and Failure Cases

- Ambulance partially occluded or visible only at night; very small crops at 96×96 pixels.
- Non-emergency ambulance or ambulance-like vehicle causing a false positive.
- Ambulance missed or detected too late to provide priority (the 39 false negatives are the current evidence).
- Two emergency vehicles approaching from conflicting directions.
- Queue spillback blocking an upstream intersection.
- Emergency route not mapped to the expected approach.
- A low-volume approach accumulating a long wait while the main road remains busy.
- Loss of connection or stale state between modules (FastAPI/MQTT).

7.7 Defects and Corrective Actions

Defects are recorded with ID, severity, reproduction steps, expected and actual behavior, root cause, corrective action, and retest evidence. Known weakness so far: ambulance recall of

91.0% on validation. Planned corrective actions: threshold tuning on the validation PR curve, oversampling or targeted augmentation of the ambulance class, larger input size, and a larger model. Each change will be tested one at a time and compared with run `cls_v4-2`.

7.8 Current Validation Status

Validated: the recognition model on the validation split (98.2% accuracy, 95.6% precision, 91.0% recall). Working but not yet quantitatively evaluated: SUMO baselines and the dashboard. Not yet validated: test-split performance, detector latency, tracking, preemption, recovery, fairness, and integrated operation.

Chapter 8

Project Management and Development Evidence

8.1 Work Completed

- Project direction, literature review, gap analysis, requirements, architecture, and evaluation metrics (Chapters 1–5).
- SUMO toolkit: 8 intersection types, 5 vehicle classes, 4 demand scenarios, fixed-time and actuated baselines.
- YOLO26 ambulance recognition model trained and validated on a 32,632-image dataset.
- Live dashboard displaying ambulance detections on video.
- Repository organized into vision, marl, sumo, comm, dashboard, and docs.

8.2 Work Remaining

- Formal detector evaluation on the test split (precision, recall, F1) and latency; improve ambulance recall.
- ByteTrack tracking and route-aware preemption with safe phase transitions.
- Fairness-aware MARL control and the recovery policy; fairness and wait metrics.
- FastAPI/MQTT integration and dashboard connection to the live system.
- Repeated experiments against baselines, analysis, and visual evidence.
- Dataset licence documentation, final report, repository documentation, and demonstration.

8.3 Individual Responsibilities

Table 8.1: Individual responsibilities

Member	Assigned work-stream	Evidence / deliverable
Abdul Hadi Ayaz Abbasi	Research, requirements and dataset	Literature review, requirements, citations; dataset preparation for model training.
Muhammad Adeel	Detection (model training)	Model training and evaluation (vision).
Marvan Shahid	Simulation and control	All simulation work: SUMO scenarios, controllers, evaluation logs (sumo).

8.4 Contribution of Each Team Member

Contributions are stated by deliverable ownership as agreed within the team. Marvan Shahid carried out all of the simulation work, Muhammad Adeel trained the ambulance recognition model, and Abdul Hadi Ayaz Abbasi prepared the dataset used for model training. Each member's part is listed in Table 8.1 and summarised with its evidence in Appendix D.

8.5 Repository and Version Control Evidence

The repository is organized as vision, marl, sumo, comm, dashboard, and docs (Table 6.2, Figure 6.4). The repository URL and commit history are given in Appendix D.

8.6 Milestones and Commit Evidence

Table 8.2: Milestones and evidence

Milestone	Evidence	Status
SUMO toolkit and baselines	Figure 6.1; configuration and run logs.	**Done**
Dataset and ambulance recognition model	Table 5.3; Figures 7.1–7.3.	**Done**
Live detection dashboard	Figure 6.3.	**Done**
Modular repository	Figure 6.4.	**Done**
Tracking and route-aware preemption	Demo recording, logs, tests.	**Iteration 2**
Recovery and fairness-aware MARL	Experiment scripts and results.	**Iteration 2**
Integrated demonstration	Tagged commit, test summary, demonstration video.	**Iteration 2**

8.7 Tools and Tool-Assisted Development

Any AI-assisted coding, code generation, annotation, literature discovery, or debugging tools used by the team are recorded in accordance with university policy. Team members remain responsible for verifying code, citations, data licences, and all submitted claims.

8.8 External Components and Dependencies

External components: SUMO, Ultralytics YOLO26 with pretrained weights, ByteTrack, FastAPI, an MQTT broker/client library, Python packages, and the ambulance/other-vehicle dataset. For each component the version, licence, source, purpose, and modifications will be recorded in docs. Dataset images are not assumed to be redistributable merely because they were downloadable.

8.9 Risks and Mitigation

Table 8.3: Risks and mitigation

Risk	Likelihood / impact	Mitigation
Missed ambulances (recall 91.0% on validation)	High / High	Threshold tuning, class balancing/augmentation, larger input/model; temporal confirmation with ByteTrack.
False-positive ambulance detection (18 on validation)	Medium / High	Confidence threshold, temporal confirmation, hard-negative tests, fail-safe control.
Class imbalance and split shift	Medium / Medium	Report per-class metrics; tune thresholds on validation only; document prevalence differences.
MARL training exceeds time/compute budget	Medium / High	Use working baselines; limit network size; fall back to heuristic recovery.
Integration delays (ByteTrack, FastAPI, MQTT)	Medium / High	Stable interfaces, incremental integration, regression tests.
Preemption increases ordinary-traffic delay	High / High	Measure affected approaches; test recovery and fairness policies.
Simulation does not represent local traffic	Medium / Medium	Document assumptions; use locally relevant vehicle classes (already in the 5 classes).
Unreproducible results	Medium / Medium	Version configs, record seeds and versions, keep raw logs.

8.10 Security and Privacy Considerations

Use public or authorized datasets and record their licences. If footage contains identifiable people or registration plates, minimize collection and retention and follow institutional and legal requirements. Do not connect the academic prototype to live traffic-control infrastructure

without authorization and safety review. Secure the FastAPI and MQTT interfaces before any use beyond a local demonstration.

8.11 Current Limitations

- Detector results are for the validation split only; test-split performance and latency are not yet measured.
- The recognition model works on small (96×96) crops with two classes and does not indicate whether an ambulance is on an emergency run.
- The dataset combines several sources, including crops taken from images and videos, so image quality and labelling vary. The splits were made randomly rather than per source, so similar crops from one video can appear in both training and validation or test data and may inflate the reported results; a source-level split is planned for FYP-2.
- Tracking, preemption, recovery, MARL, and integration are not yet implemented.
- No traffic-performance results exist yet; they will depend on road geometry, demand, driver behavior, and scenario assumptions.
- A fairness objective can improve measured equity but does not give a formal maximum-wait guarantee.
- The system is not intended for live-road deployment in its academic prototype form.

Chapter 9

Remaining FYP Plan

9.1 Remaining Scope

We keep the combined scope while it remains feasible. A complete tracked-detection, preemption, and recovery prototype with rigorous SUMO evaluation, even with a rule-based recovery policy, is preferable to an incomplete MARL system.

9.2 Remaining Modules and Features

Table 9.1: Remaining deliverables by module

Module	Remaining deliverable
M1 – Perception	Test-split evaluation, latency, recall improvement, ByteTrack tracking, error analysis.
M2 – State estimation	Queue/wait/phase state and approach mapping.
M3 – Dynamic control	Demand-aware timing beyond the actuated baseline; empty-green handling.
M4 – Emergency priority	Validated priority event, route-aware preemption, safe phase transitions, corridor coordination.
M5 – Fair recovery / MARL	Queue/wait-aware recovery, fairness-aware MARL, maximum-wait monitoring.
M6 – Evaluation	Baseline comparisons, repeated runs, plots, result analysis.
Integration	FastAPI/MQTT interfaces and dashboard connection.

9.3 Remaining Iterations

Table 9.2: Remaining iterations

Iteration	Main tasks	Exit criterion
Iteration 1: Foundation (**Done**)	Sensing and simulation foundations; repository.	Trained model, dashboard, SUMO baselines.
Iteration 2: Tracking and control	ByteTrack, route-aware preemption, recovery, fairness-aware MARL.	A scripted ambulance scenario triggers and completes a logged preemption with recovery.
Iteration 2: Integration	FastAPI, MQTT, dashboard connection.	End-to-end run from video detection to signal action in SUMO.
Iteration 3: Evaluation and packaging	Repeated seeds, sensitivity analysis, documentation, demonstration.	Reproducible experiment package and final report.

9.4 Implementation Plan

- Complete the detector evaluation (test split, latency) and iterate on recall.
- Add ByteTrack and define the detection-event interface (confidence, timestamp, approach, track ID).
- Implement route-aware preemption on top of the SUMO baselines and verify safety constraints with automated tests.
- Implement recovery using accumulated waiting time and queue pressure; define fairness metrics before tuning.
- Train the fairness-aware MARL controller and compare it with the rule-based policy on the same scenarios.
- Connect modules through FastAPI and MQTT and link the dashboard.
- Package code, configurations, logs, model weights (where licensing permits), and a reproducible README.

9.5 Testing and Validation Plan

Use a fixed scenario matrix and repeated random seeds. Report detection metrics on the test split, then traffic metrics in SUMO. For integrated testing, introduce controlled detection delay, missed detections, and false alarms to measure how perception uncertainty affects control. Report means and variability; avoid broad claims from a single run.

9.6 Deployment Plan

The deliverable is a research prototype for offline simulation and recorded-video demonstration. A live camera feed may be shown only as a non-controlling dashboard demonstration. No real traffic signals are actuated. Operational deployment would require hardware integration, field validation, fail-safe design, cybersecurity review, regulatory approval, and collaboration with traffic authorities.

9.7 Tentative Timeline

This is a relative schedule and will be coordinated with the official FYP calendar.

Table 9.3: Tentative timeline for FYP-2

Period	Planned work	Deliverable
Weeks 1–2	Test-split and latency evaluation; recall improvement; ByteTrack integration.	Detector report; tracked detections.
Weeks 3–4	Detection-event interface; route-aware preemption in SUMO.	Working priority demonstration.
Weeks 5–6	Recovery policy and fairness/wait metrics.	Recovery comparison; metric scripts.
Weeks 7–8	Fairness-aware MARL training and tuning.	Trained policy; training logs.
Weeks 9–10	FastAPI/MQTT integration; dashboard connection; scenario matrix runs.	Integrated prototype; results tables and plots.
Weeks 11–12	Sensitivity analysis, documentation, defense preparation.	Final prototype, reproducibility package, report.

9.8 Expected Final Deliverables

- A documented, appropriately licensed ambulance dataset with class mapping and split manifests.
- A trained and evaluated recognition/detection model with tracking.
- SUMO networks, scenarios, and baseline controllers (delivered in Iteration 1).
- Dynamic signal controller with emergency priority and post-preemption recovery.
- Experiment scripts for baselines and the proposed policy.
- Quantitative results, plots, limitations, and failure-case analysis.
- Versioned repository, setup instructions, test evidence, and a demonstration (dashboard).
- Final report with verified references and authentic progress evidence.

Chapter 10

Conclusion

This report defines an AI-based dynamic traffic-management project that integrates ambulance detection, emergency signal priority, adaptive traffic control, and fairness-aware recovery. The motivation is the separation between work that improves emergency-vehicle passage and work that improves fairness for ordinary traffic. The evaluation therefore measures both ambulance response and the effect of preemption on other road users, including waiting-time tails and fairness indicators.

Iteration 1 has delivered the sensing and simulation foundations: a SUMO toolkit with 8 intersection types, 5 vehicle classes, 4 demand scenarios and fixed-time and actuated baselines; a YOLO26 ambulance recognition model that reaches 98.2% validation accuracy with 95.6% ambulance precision and 91.0% recall; a live detection dashboard; and a modular repository. The main technical finding is that ambulance recall, not overall accuracy, is the weak point, which matters because missed ambulances are the costlier error.

The remaining work, planned for Iteration 2 in FYP-2, is ByteTrack tracking, route-aware preemption, fairness-aware MARL control, and FastAPI/MQTT integration, followed by repeated SUMO experiments against the baselines. The report makes no traffic-performance claims yet; these will be added once measured.

Appendix A

Proposal Action Register

Table A.1: Detailed proposal action register

No.	Action	Owner	Due	Evidence / status
A1	Confirm final scope and measurable acceptance criteria.	Project team	FYP-2	Scope and criteria record
A2	Verify literature details and complete references.	Project team	FYP-2	Verified reference list
A3	Document dataset sources and licences.	Project team	FYP-2	**Done**; Table A.2 (sources and licences)
A4	Evaluate the detector on the test split and measure latency.	Vision	Iteration 2	Metrics report
A5	Build reproducible SUMO baseline.	Simulation	Iteration 1	**Done**; configuration and run log
A6	Implement emergency priority and recovery.	Project team	Iteration 2	Implementation and tests
A7	Run comparative evaluation and document results.	Project team	FYP-2	Raw logs, plots, analysis

Table A.2: Dataset sources and licences

No.	Source (Roboflow Universe)	Workspace / project (version)	Licence	Classes as released
S1	Pakistani Vehicle Detection YOLO	baakhs-workspace / pakistani-vehicle-detection-yolo-h5oie (v1)	CC BY 4.0	car, emv, htv, motorcycle, rikshaw
S2	Ambulance	ambulance-uj5pi / ambulance-sjpea (v20)	Public Domain	ambulance, auto, bike, bus, car, siren
S3	Ambulance Detection	ambulance-b8kgx / ambulance-detection-tnzha (v3)	CC BY 4.0	ambulance, siren
S4	Ambulance	srivalli-yada / ambulance-wmxl5 (v5)	CC BY 4.0	ambulance
S5	Detecting the Ambulance	traffic-analysis-wl8wg / detecting-the-ambulance (v4)	CC BY 4.0	ambulance
S6	Vehicles-OpenImages	roboflow-gw7yv / vehicles-openimages (v1)	CC BY 4.0	Ambulance, Bus, Car, Motorcycle, Truck

All sources were obtained from Roboflow Universe. Five are licensed CC BY 4.0, so attribution to the original authors is required; S2 is released as Public Domain. Class names are those in each exported data file; the final dataset uses only the two classes `ambulance` and `other_vehicle`. Source links:

S1: `https://universe.roboflow.com/baakhs-workspace/pakistani-vehicle-detection-yolo-h5oie/dataset/1`

S2: `https://universe.roboflow.com/ambulance-uj5pi/ambulance-sjpea/dataset/20`

S3: `https://universe.roboflow.com/ambulance-b8kgx/ambulance-detection-tnzha/dataset/3`

S4: `https://universe.roboflow.com/srivalli-yada/ambulance-wmxl5/dataset/5`

S5: `https://universe.roboflow.com/traffic-analysis-wl8wg/detecting-the-ambulance/dataset/4`

S6: `https://universe.roboflow.com/roboflow-gw7yv/vehicles-openimages/dataset/1`

Table A.3: Supervisor feedback and team actions

No.	Date	Supervisor feedback (Dr. Suleman Khan)	Action taken by the team
1	09-09-2026	Idea discussion. The supervisor asked the team to review papers related to the project idea.	Reviewed related work on adaptive signal control, emergency-vehicle preemption and vehicle detection; used for the literature review (Chapter 2).
2	16-09-2026	Proposal discussion. The project scope needed to be refined and additional modules needed to be added.	Scope revised and the system modules (sensing, simulation, tracking and control, integration) added to the plan.
3	23-09-2026	Data preprocessing and model discussion. Dataset augmentation and preprocessing were discussed; a dashboard showing live updates needs to be prepared.	Augmentation and preprocessing applied to the classifier dataset. Dashboard with video view built; live-update connection planned for Iteration 2.
4	07-10-2026	Guidance on the mid evaluation.	Mid-evaluation report and demonstration prepared.
5	14-10-2026	Models and simulation evaluation. Multiple real Pakistani traffic scenarios need to be added to the simulation.	Simulation currently has 8 intersection types, 5 vehicle classes and 4 demand scenarios; further Pakistan-specific scenarios to be added in Iteration 2.

Appendix B

Detailed Requirements

Table B.1: Detailed requirements

ID	Requirement statement	Verification artifact
FR-01	Accept supported image/video input or simulated events.	Input validation test.
FR-02	Return ambulance class/confidence (and boxes in detection mode).	Held-out evaluation.
FR-03	Timestamp and log detection events.	Log inspection.
FR-04	Track phase, queues, waits, and emergency approach.	State-estimation tests.
FR-05	Handle missing/stale/low-confidence detections safely.	Negative and stale-input tests.
FR-06	Enforce configured phase and duration constraints.	Signal-state tests.
FR-07	Shorten empty green only when safe and permitted.	Empty-approach scenario.
FR-08	Request priority for a validated approaching ambulance.	Emergency-trip scenario.
FR-09	Record preemption and recovery lifecycle events.	Event-log review.
FR-10	Allocate recovery service using queue/wait information.	Recovery comparison.
FR-11	Compare with fixed-time and adaptive baselines.	Result table and scripts.
FR-12	Export configuration and metrics for reproduction.	Re-run by another member.

Appendix C

Detailed Test Cases

Table C.1: Detailed test cases

ID	Inputs / setup	Procedure	Expected result
TC-01	Labelled test images	Run model; compare with labels.	Precision/recall/F1 and errors recorded.
TC-02	No-ambulance images/video	Run at configured threshold.	No unverified priority event.
TC-03	Stale detection event	Replay event older than allowed.	Priority request rejected or expires.
TC-04	Empty green approach	Run demand-responsive controller.	Phase may advance after safety checks.
TC-05	Ambulance approaching red	Trigger emergency event.	Compatible phase with valid clearance.
TC-06	Ambulance clears intersection	Complete emergency trip.	Preemption clears; recovery starts.
TC-07	One approach with excessive wait	Run recovery under competing queues.	Wait-aware rule activated.
TC-08	Conflicting phase request	Issue an incompatible command.	Unsafe simultaneous green prevented.
TC-09	No emergency event	Run baseline scenario.	Controller functions; metrics logged.
TC-10	Same scenario and seed	Repeat run.	Results match within tolerance.

Appendix D

Repository and Contribution Evidence

The repository is organized into the following top-level modules (see Table 6.2 and Figure 6.4):

`vision/`	ambulance recognition model, training, inference
`sumo/`	networks, scenarios, demand, baseline controllers
`marl/`	fairness-aware multi-agent RL controller
`comm/`	FastAPI / MQTT communication
`dashboard/`	live ambulance detection dashboard
`docs/`	documentation

Source code repository (GitHub): `https://github.com/MarvanShahid/EquiFlow`

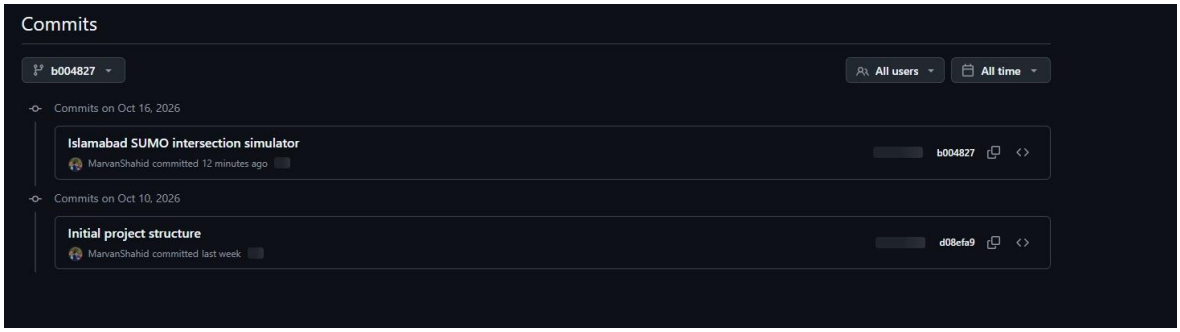


Figure D.1: Commit history of the EquiFlow repository.

Each member's contribution is summarised in Table D.1 by deliverable ownership. Marvan Shahid carried out all of the simulation work, Muhammad Adeel trained the ambulance recognition model, and Abdul Hadi Ayaz Abbasi prepared the dataset used for model training.

Table D.1: Contribution of each team member

Member	Workstream	Contribution	Evidence
Abdul Hadi Ayaz Abbasi (23I-0707)	Research, requirements and dataset	Prepared the dataset used to train the ambulance recognition model; literature review, requirements and citations.	Dataset counts (Table 5.3); sources and licences (Table A.2).
Muhammad Adeel (23I-0739)	Model training	Trained and evaluated the YOLO26 ambulance recognition model.	`vision/` module; training run `cls_v4-2`, confusion matrix and training curves (Chapter 7).
Marvan Shahid (23I-0800)	Simulation	All simulation work: SUMO networks, scenarios, demand and baseline controllers, with run and evaluation logs.	`sumo/` module; SUMO toolkit figure (Figure 6.1); GitHub repository.

Credentials, access tokens, and private data are not included.

Appendix E

Additional Design Diagrams

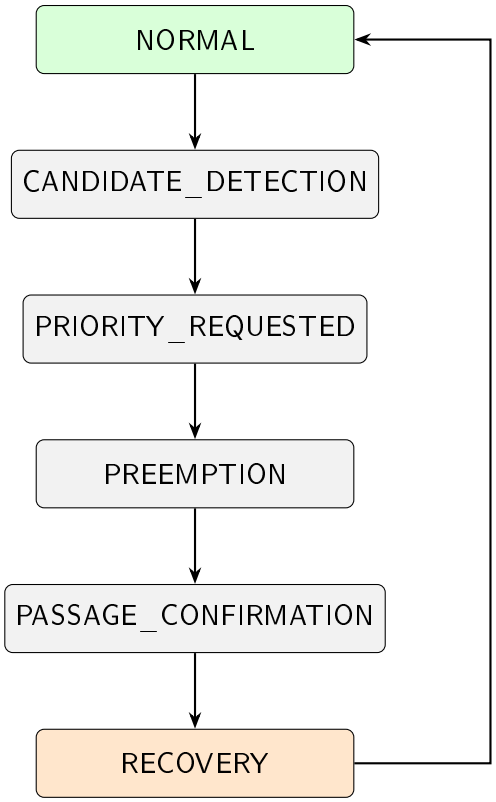


Figure E.1: Controller state machine for emergency handling.

The following supplementary diagrams complete the design documentation: the system architecture (Figure E.2), the activity diagram of normal operation, preemption and recovery (Figure E.3), the sequence diagram of an ambulance passing two intersections (Figure E.4), and the data-flow diagram (Figure E.5).

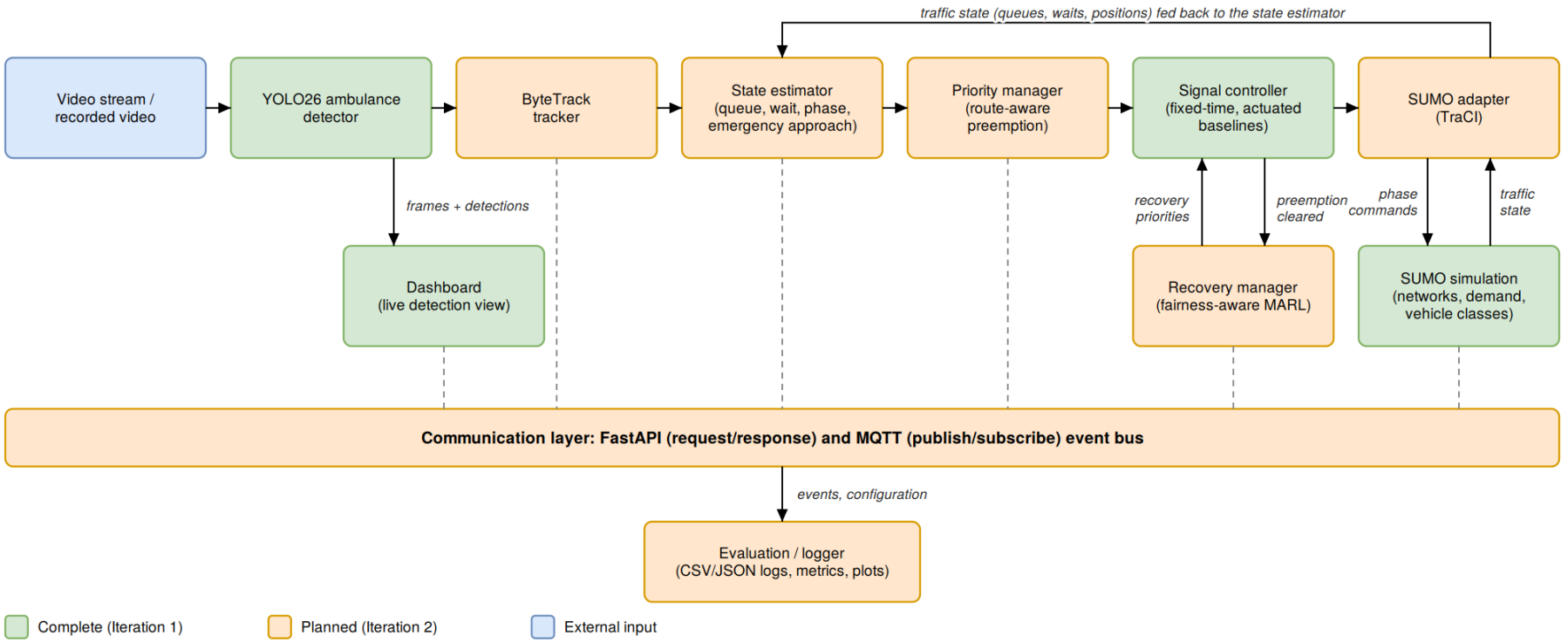


Figure E.2: System architecture: modules, data flows, and implementation status (green: complete in Iteration 1; orange: planned for Iteration 2; blue: external input).

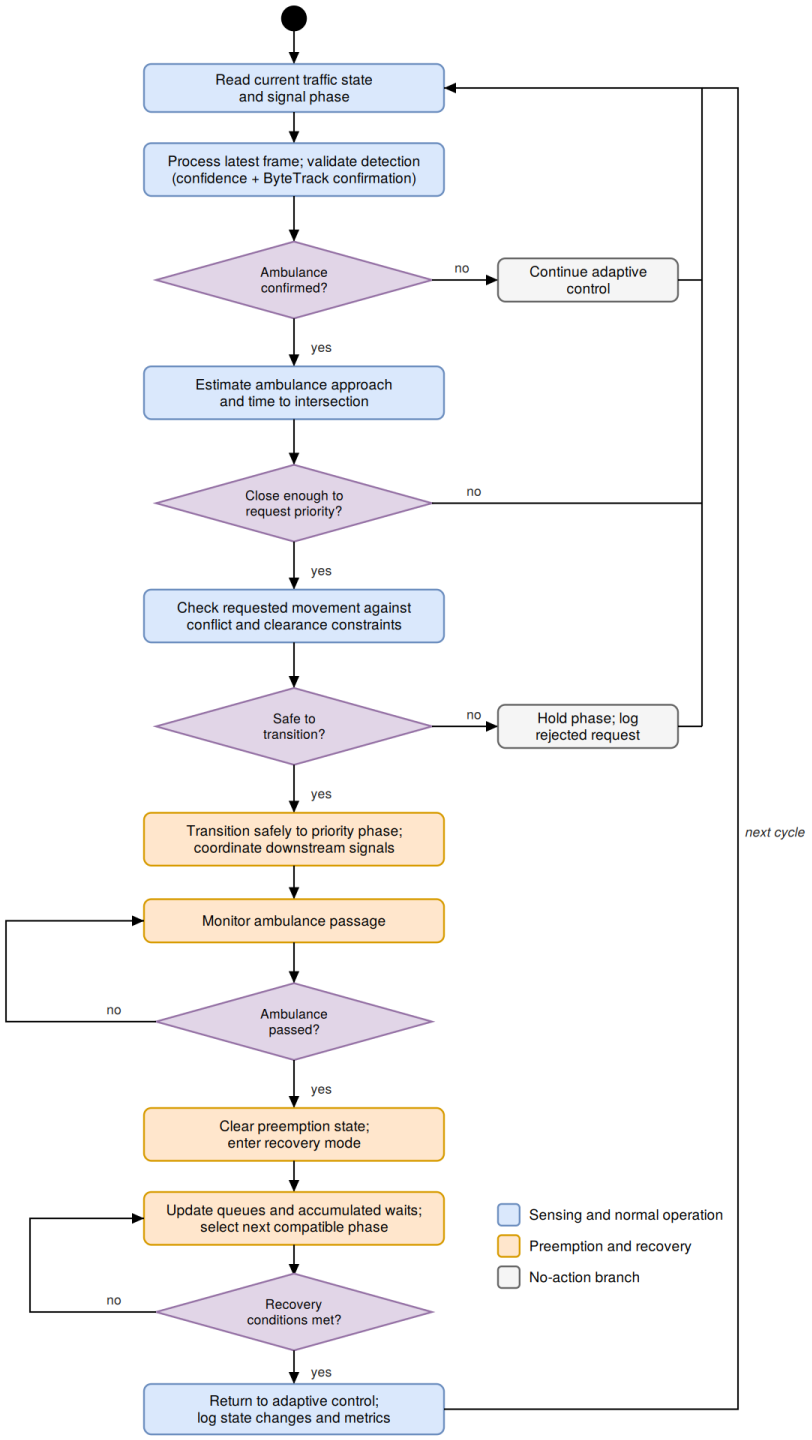


Figure E.3: Activity diagram: normal operation, detection, safe preemption, passage, recovery, and return to normal.

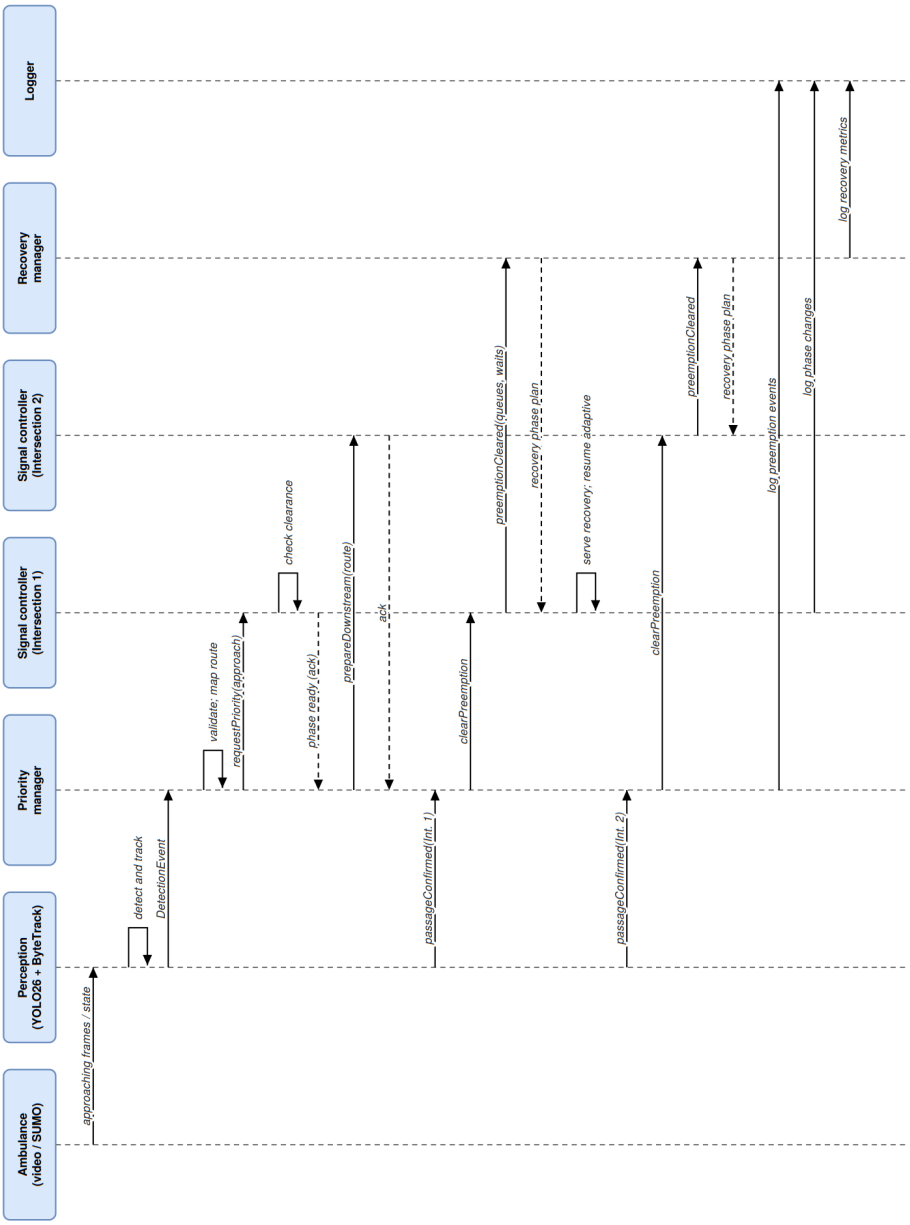


Figure E.4: Sequence diagram: an ambulance approaching a sequence of two intersections.

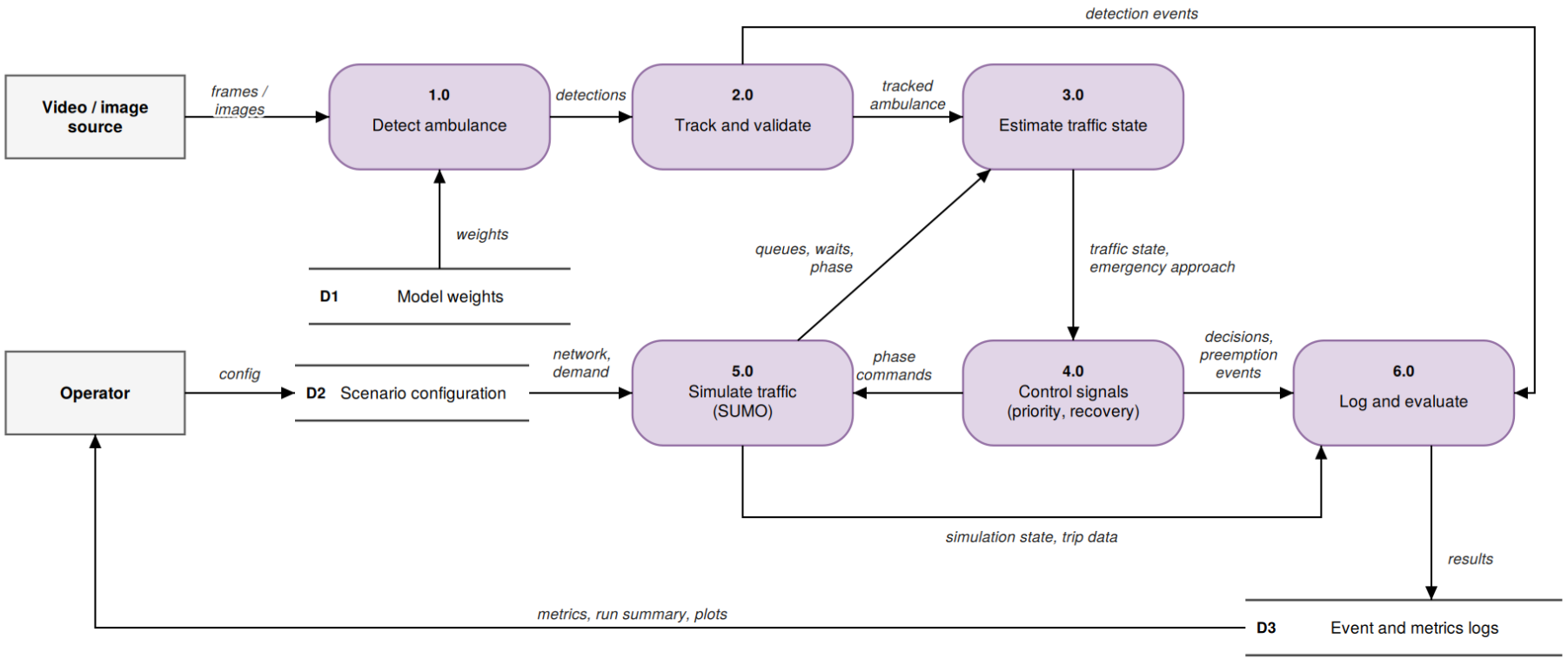


Figure E.5: Data-flow diagram: images, model weights, simulation state, controller actions, and metric outputs.

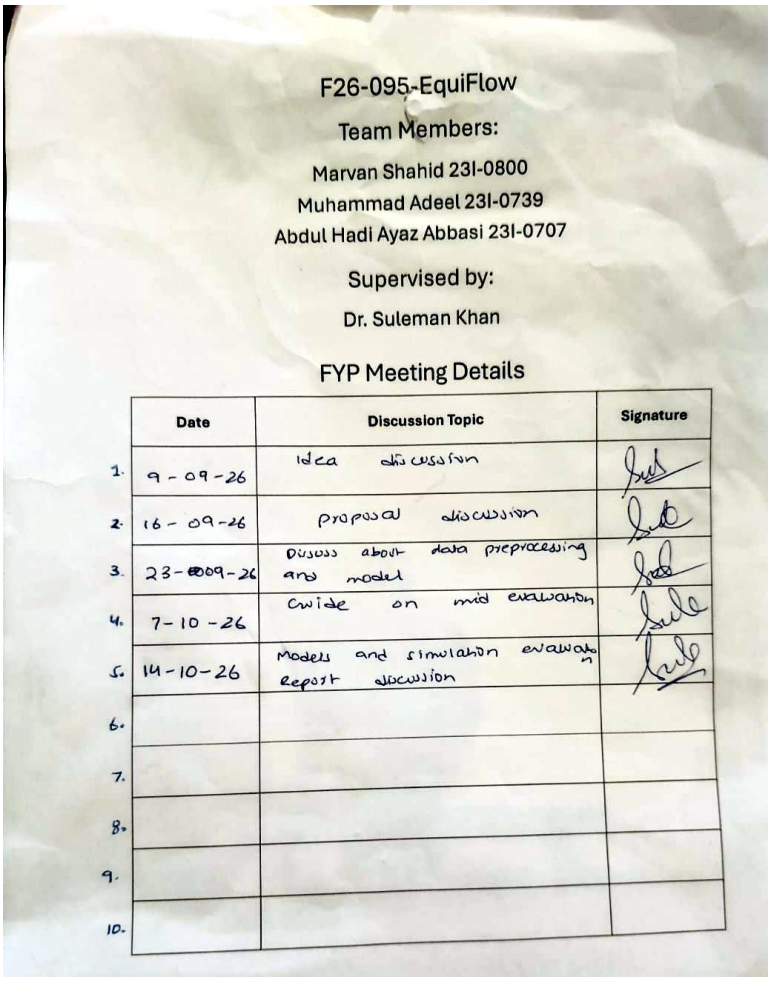
Appendix F

Meeting and Progress Evidence

The team held regular supervisory meetings with Dr. Suleman Khan. The signed meeting log (Figure F.1) records five meetings to date; Table F.1 summarises them.

Table F.1: Supervisory meeting log (as recorded on the signed sheet)

No.	Date	Discussion topic
1	09-09-2026	Idea discussion
2	16-09-2026	Proposal discussion
3	23-09-2026	Data preprocessing and model discussion
4	07-10-2026	Guidance on mid evaluation
5	14-10-2026	Models and simulation evaluation; report discussion



F26-095-EquiFlow

Team Members:

Marvan Shahid 23I-0800
Muhammad Adeel 23I-0739
Abdul Hadi Ayaz Abbasi 23I-0707

Supervised by:

Dr. Suleman Khan

FYP Meeting Details

	Date	Discussion Topic	Signature
1.	9-09-26	idea discussion	
2.	16-09-26	proposal discussion	
3.	23-09-26	Discuss about data preprocessing and model	
4.	7-10-26	Guide on mid evaluation	
5.	14-10-26	Models and simulation evaluation report discussion	
6.			
7.			
8.			
9.			
10.			

Figure F.1: Signed FYP meeting log (F26-095-EquiFlow), supervised by Dr. Suleman Khan.

Bibliography

[1] Fang, W., Zhao, X., & Zhang, C. (2024). Fairness-aware multi-agent reinforcement learning and visual perception for adaptive traffic signal control (FA-MAPPO). *Optoelectronics Letters*, 20(12), 764. https://doi.org/10.1007/s11801-024-3267-2

[2] Haydari, A., Aggarwal, V., Zhang, M., & Chuah, C.-N. (2024). Constrained Reinforcement Learning for Fair and Environmentally Efficient Traffic Signal Controllers (FAirLight). *Journal on Autonomous Transportation Systems*. https://doi.org/10.1145/3676169

[3] Malik, M. Z., Nazir, S., & Khan, H. U. (2023). AI-based System on Enhancing the Capabilities of Transport System: A Systemic Literature Review. *IEEE ISIEA 2023*. https://doi.org/10.1109/ISIEA58478.2023.10212340

[4] Raeis, M., & Leon-Garcia, A. (2021). A Deep Reinforcement Learning Approach for Fair Traffic Signal Control. arXiv preprint. https://arxiv.org/abs/2107.10146

[5] Ultralytics. (n.d.). Ultralytics YOLO documentation. https://docs.ultralytics.com/

[6] Eclipse SUMO. (n.d.). SUMO documentation. https://sumo.dlr.de/docs/

[7] Su, H., Zhong, Y. D., Dey, B., & Chakraborty, A. (2022). EMVLight: A decentralized reinforcement learning framework for efficient passage of emergency vehicles. *Proceedings of the AAAI Conference on Artificial Intelligence*, 36(4), 4593–4601.

[8] Mecocci, A., & Grassi, C. (2024). RTAIAED: A real-time ambulance in an emergency detector with a pyramidal part-based model composed of MFCCs and YOLOv8. *Sensors*, 24(7), 2321.

[9] Wan, Y., Wu, K., Shi, T., & Wang, J. (2024). Fair and efficient traffic light control with reinforcement learning. In *Proceedings of ISICN 2024*. Springer, pp. 17–33.

[10] Pioro, M. (2025). FairSignal: A multi-agent reinforcement learning approach considering fairness for multi-intersection traffic signal control. IEEE. https://ieeexplore.ieee.org/abstract/document/10944711

[11] Zhang, Y., Sun, P., Jiang, Y., Yu, D., Weng, F., Yuan, Z., Luo, P., Liu, W., & Wang, X. (2022). ByteTrack: Multi-Object Tracking by Associating Every Detection Box.

European Conference on Computer Vision (ECCV). https://arxiv.org/abs/2110.06864